

Certifying Γ -Robust Discrete Pricing via Full-Breakpoint Multiple-Choice Knapsack Decomposition

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Abstract

We study Γ -robust finite-menu multiple-choice knapsack problems motivated by discrete pricing with robust margin constraints. For finite menus and integer Γ , the Bertsimas–Sim protection term yields an exact finite θ decomposition over the full original breakpoint set. For each fixed θ , the problem is an ordinary MCKP whose LP relaxation is solved exactly by classical upper-hull greedy geometry and has an optimal solution with at most one fractionally mixed class. This structure gives HullRound, a robust-feasible primal method with a one-item additive certificate; an exact θ -enumerated branch-and-bound procedure that preserves original non-dominated options and reports valid gaps under limits; and a full-breakpoint parametric sweep with safe reuse/rebuild diagnostics. Experiments validate the certificates and sweep equivalence, show near-exact HullRound primal quality on certified non-tight rows, and identify tight-capacity rows as the main exact-solver bottleneck. Generic MILP solvers remain strong runtime baselines, while the proposed decomposition provides explicit certificates, gap semantics, and diagnostic structure not exposed by black-box formulations. A controlled semi-synthetic pricing study illustrates the revenue, protection, and price-movement tradeoffs induced by Γ .

Keywords: Discrete Optimization; Robust Optimization; Pricing; Knapsack Problem

1 Introduction

Finite-menu pricing and other discrete portfolio decisions often combine a nominal objective with a margin or service constraint that must remain valid under parameter error. In pricing, each product may have only a small admissible menu because of reference-price rules, regulatory constraints, psychological price endings, or operational price-movement limits. The same finite-menu structure appears more broadly whenever a decision maker chooses one option from each group under a portfolio-level resource or margin requirement.

After discretization and a baseline–slack transformation, the deterministic core is a Multiple-Choice Knapsack Problem (MCKP). The robust extension considered here protects the capacity/margin side against Γ -budget uncertainty. The pricing interpretation starts from a ratio-margin constraint $N(\mathbf{x})/D(\mathbf{x}) \geq \Delta$; this ratio interpretation requires a strictly positive realized denominator over the uncertainty set. Algebraically, the robust model uses the residual inequality $\tilde{N}(\mathbf{x}) - \Delta\tilde{D}(\mathbf{x}) \geq 0$ as the primitive feasibility condition.

The algorithmic difficulty is that Γ -budget coupling destroys the itemwise separability that makes ordinary finite-menu problems tractable after relaxation. A generic robust MILP formulation is available, but it obscures which solutions are certified by MCKP structure, which gaps are valid under limits, and when rounding is intrinsically safe. This paper exposes the robust-MCKP structure directly and uses it to build auditable primal solutions, bounds, and diagnostics.

The key observation is that the Bertsimas–Sim protection term has a scalar θ representation. Because the menus are finite, exact global optimization only needs the finite breakpoint set

$$\mathcal{B} = \{0\} \cup \{t_{ij} \mid \text{all original admissible options } (i, j)\}.$$

For a fixed θ , the robust problem is again an ordinary MCKP. Its LP relaxation has the classical upper-hull greedy structure and admits an optimal solution with at most one fractionally mixed item, even under ties in segment density. This converts robust coupling into a family of structured deterministic subproblems and yields both primal rounding certificates and exact anytime bounds.

Contributions. The paper is a robust-MCKP algorithmic study, with discrete pricing serving as the motivating application and source of structured instances.

- (i) Exact finite-menu reduction and full-breakpoint θ decomposition for integer-budget Bertsimas–Sim robust MCKP.
- (ii) Fixed- θ upper-hull LP oracle and HullRound certificate: round-down is robust-feasible and loses at most one item jump.
- (iii) Full- θ exact branch-and-bound with valid LP-bound pruning, valid anytime gaps, and exact search over original non-dominated options, including below-hull options.
- (iv) Full-breakpoint parametric sweep with safe item-local hull reuse only when the induced cost–value point set is unchanged.
- (v) Empirical certification evidence: near-exact HullRound quality on non-tight rows, tight-capacity rows as the bottleneck, and SCIP/HiGHS as strong median-runtime baselines.

1.1 Main result and certification objects

The paper’s main claim is a certification theorem and algorithmic framework. For finite menus and integer Γ , robust optimization is exactly equivalent to maximizing over the full original breakpoint set $\mathcal{B}$. Each fixed- θ problem is an ordinary MCKP whose LP relaxation has an explicit upper-hull greedy optimum with at most one fractionally mixed class. Rounding that class down gives a robust-feasible incumbent with a one-item additive loss certificate; solving or validly pruning all fixed- θ subproblems gives an exact global optimum, and under limits gives a valid global gap once every θ has an initialized upper-bound record. The exact layer searches over original non-dominated options, including below-hull options that are irrelevant only to LP upper bounds.

This statement combines classical ingredients without claiming them as new. Bertsimas–Sim budgeted uncertainty supplies the scalar protection representation; ordinary MCKP supplies upper-hull LP geometry and one-fractional-class structure; generic robust MILP supplies a strong baseline. The contribution is the certifying robust-MCKP synthesis: full original-breakpoint θ enumeration, fixed- θ hull-greedy LP bounds, robust-feasible one-item rounding certificates, below-hull-preserving exact search, valid gap accounting, and safe full-breakpoint sweep diagnostics. The detailed contrast with generic robust MILP appears in [Table 1](#).¹

1.2 Scope and standing assumptions

The standing model has finite menus and exactly one admissible option selected from each class. In the pricing interpretation, reference prices satisfy $a_i > 0$, admissible price menus are nonnegative after the reference-band restriction, exposures satisfy $\omega_i \geq 0$, and nominal and realized demands are nonnegative on the admissible menus. The ratio-margin interpretation additionally requires the realized denominator $\tilde{D}(\mathbf{x}) >$

¹An earlier arXiv working-paper version ([Shao, 2026](#)) studied robust discrete pricing through MCKP reductions and reported preliminary synthetic and stylized pricing experiments. The present manuscript supersedes that version by reframing the contribution around robust-MCKP certification, full original-breakpoint θ decomposition, below-hull-safe exact search, HullRound certificates, valid gap semantics, sweep diagnostics, and a controlled robustness-design pricing illustration.

0 for every admissible price vector and every uncertainty realization; if this positive-volume condition is not imposed, the paper's algorithms apply to the additive residual constraint $\tilde{N}(\mathbf{x}) - \Delta \tilde{D}(\mathbf{x}) \geq 0$, which is the algebraic primitive used throughout. The objective is nominal revenue under robust feasibility, reflecting a decision maker who wants worst-case margin protection without optimizing worst-case revenue.

The exact theory and implementation in this paper handle integer cardinality budgets $\Gamma \in \{0, \dots, n\}$. Fractional Bertsimas–Sim budgets are standard, but would add a fractional next-largest protection term and require a separate finite-menu treatment; they are outside the present implementation. The pricing application is own-price and separable, and the experiments are synthetic or semi-synthetic rather than scanner-data validation. Finally, upper hulls are LP-bound objects. Exact integer search preserves original non-dominated options, including below-hull options that may be optimal or necessary for feasibility.

1.3 Problem Formulation

For each class $i \in \{1, \dots, n\}$, choose one price level $x_i \in \mathcal{X}_i = \{x_{i,1}, \dots, x_{i,m}\}$. The data are a reference price $a_i > 0$, exposure weight ω_i , nominal demand $\hat{g}_i : \mathcal{X}_i \rightarrow \mathbb{R}_{\geq 0}$, relative tolerance σ_i , and margin target $\Delta > 0$. We assume the demand functions $\hat{g}_i$ depend only on the own-price x_i and not on other items' prices, which is appropriate when cross-product substitution is negligible. This is a modelling restriction; the semi-synthetic study in [Section 8](#) is therefore interpreted as a robustness-design illustration rather than transaction-level pricing validation.

Throughout the admissible menus, assume $a_i > 0$, $\omega_i \geq 0$, $\hat{g}_i(x) \geq 0$, and $D(\mathbf{x}) > 0$ for every feasible price vector.

$$N(\mathbf{x}) := \sum_{i=1}^n \omega_i x_i \hat{g}_i(x_i), \quad D(\mathbf{x}) := \sum_{i=1}^n \omega_i a_i \hat{g}_i(x_i) > 0, \quad S(\mathbf{x}) := N(\mathbf{x}) - \Delta \cdot D(\mathbf{x}).$$

The nominal finite-menu problem is

$$\max_{\mathbf{x}} \quad N(\mathbf{x}) = \sum_{i=1}^n \omega_i x_i \hat{g}_i(x_i), \tag{1}$$

$$\text{s. t.} \quad \frac{N(\mathbf{x})}{D(\mathbf{x})} \geq \Delta \Leftrightarrow S(\mathbf{x}) \geq 0, \tag{2}$$

$$\frac{|x_i - a_i|}{a_i} \leq \sigma_i \Leftrightarrow x_i \in [(1 - \sigma_i)a_i, (1 + \sigma_i)a_i] \quad (\forall i). \tag{3}$$

The second constraint can be enforced by defining the admissible menu

$$I_i := [(1 - \sigma_i)a_i, (1 + \sigma_i)a_i], \quad \mathcal{X}_i^{\text{ad}} := \mathcal{X}_i \cap I_i \neq \emptyset, \quad i = 1, \dots, n,$$

and henceforth restrict all decisions to $x_i \in \mathcal{X}_i^{\text{ad}}$.

1.4 Reduction to MCKP

Introduce binary selection variables $z_{ij} \in \{0, 1\}$ indicating the choice $x_i = x_{i,j}$. For each option $x_{i,j} \in \mathcal{X}_i$, define

$$v_{ij} := \omega_i x_{i,j} \hat{g}_i(x_{i,j}), \tag{4}$$

$$d_{ij} := \omega_i a_i \hat{g}_i(x_{i,j}), \tag{5}$$

$$s_{ij} := v_{ij} - \Delta d_{ij}. \tag{6}$$

Let $J_i^{\text{ad}} := \{j \in \{1, \dots, m\} : x_{i,j} \in X_i^{\text{ad}}\}$ denote the admissible indices. Then

$$N(\mathbf{x}) = \sum_{i=1}^n \sum_{j \in J_i^{\text{ad}}} v_{ij} z_{ij}, \quad S(\mathbf{x}) = \sum_{i=1}^n \sum_{j \in J_i^{\text{ad}}} s_{ij} z_{ij}, \quad \sum_{j \in J_i^{\text{ad}}} z_{ij} = 1 \quad (\forall i).$$

To obtain a single knapsack constraint with nonnegative costs, apply a baseline–slack transformation. For each i choose a baseline option

$$\begin{aligned} j^*(i) &\in \arg \max_{j \in J_i^{\text{ad}}} s_{ij}, \\ S_{\max} &:= \sum_{i=1}^n s_{i j^*(i)}, \\ c_{ij} &:= s_{i j^*(i)} - s_{ij} \quad (\geq 0) \quad (j \in J_i^{\text{ad}}). \end{aligned}$$

If $S_{\max} < 0$, then even the itemwise maximum-margin choices violate the margin constraint, so the original pricing problem is infeasible; otherwise the transformed MCKP has capacity $S_{\max}$. Then

$$\sum_{i=1}^n \sum_{j \in J_i^{\text{ad}}} s_{ij} z_{ij} = \sum_{i=1}^n s_{i j^*(i)} - \sum_{i=1}^n \sum_{j \in J_i^{\text{ad}}} c_{ij} z_{ij} = S_{\max} - \sum_{i=1}^n \sum_{j \in J_i^{\text{ad}}} c_{ij} z_{ij}.$$

Hence the ratio constraint $S(\mathbf{x}) \geq 0$ is equivalent to the knapsack inequality

$$\sum_{i=1}^n \sum_{j \in J_i^{\text{ad}}} c_{ij} z_{ij} \leq S_{\max}.$$

The pricing problem is therefore exactly the multiple-choice knapsack problem (MCKP)

$$\max \quad \sum_{i=1}^n \sum_{j \in J_i^{\text{ad}}} v_{ij} z_{ij} \tag{7}$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{j \in J_i^{\text{ad}}} c_{ij} z_{ij} \leq S_{\max}, \tag{8}$$

$$\sum_{j \in J_i^{\text{ad}}} z_{ij} = 1 \quad (i = 1, \dots, n), \tag{9}$$

$$z_{ij} \in \{0, 1\} \quad (\forall i, j \in J_i^{\text{ad}}). \tag{10}$$

2 Related Work

MCKP foundations. The Multiple-Choice Knapsack Problem (MCKP) is a classical model in which exactly one option is selected from each group under a knapsack constraint. Early work established complexity and exact methods based on branch-and-bound, surrogate relaxations, and dominance preprocessing (Nauss, 1978; Ibaraki et al., 1978; Sinha and Zoltners, 1979). A central structural fact is that an optimal LP relaxation can be chosen with at most one fractionally mixed group (Dyer, 1984). This near-integrality underlies many exact and heuristic methods, including convex-hull and core-based algorithms (Pisinger, 1995; Kellerer et al., 2004). We use this geometry as a primitive rather than claiming it as a new result.

MCKP variants and matheuristics. Large finite-menu resource-allocation and pricing models often add setup decisions, side constraints, or multiple resources. Recent work on MCKP with setups illustrates how exact algorithms, local branching, and decomposition can be combined effectively for richer deterministic models (Boukhari and Hifi, 2023, 2024). Robust multidimensional variants move in a different direction by protecting against parameter uncertainty, but they are commonly handled through generic robust MILP formulations or matheuristics rather than by preserving the one-knapsack LP geometry (Caserta and Voß, 2019).

Budgeted robust optimization and robust knapsack. Robust optimization provides tractable worst-case models when distributional information is limited (Ben-Tal and Nemirovski, 1999). The budgeted uncertainty model of Bertsimas and Sim (2004) interpolates between nominal and box protection while often preserving linear reformulations. Exact and combinatorial approaches for robust knapsack variants have been studied in the single-choice setting (Monaci et al., 2013), and locally budgeted uncertainty further refines the modeling of heterogeneous uncertainty budgets (Goerigk et al., 2021). The present paper keeps the classical global Γ budget but asks how its scalar protection representation interacts with MCKP LP structure.

Robust pricing. Pricing under demand uncertainty can be formulated through stochastic, distributionally robust, or interval-robust models depending on the available data. Interval-robust multi-product price optimization protects decisions against demand-parameter misspecification without requiring a full probability law (Hamzeei et al., 2022). Our pricing model is more restricted empirically: it is own-price and finite-menu, with constructed uncertainty bands. Its role is to motivate robust-MCKP algorithms and supply controlled pricing-derived test instances.

Positioning. The gap addressed here is specialized certification structure for Γ -robust MCKP. A generic robust MILP can certify optimality and is a strong runtime baseline in our experiments, but it does not expose the fixed- θ subproblem bounds, one-item rounding loss, or below-hull integer-safety issue that drive the certification analysis. Table 1 summarizes the distinction.

Table 1. What the decomposition adds beyond a generic robust MILP formulation.

Aspect	Generic robust MILP formulation	This paper’s certifying robust-MCKP decomposition
Robust coupling	Compact counterpart; the solver handles the coupling.	Scalar θ separates the coupling into fixed- θ MCKPs over the full original breakpoint set.
LP bounds	Solver-generated relaxations and presolve.	Explicit upper-hull greedy LP oracle with one fractionally mixed class.
Rounding	No intrinsic robust-feasible rounding certificate.	HullRound round-down certificate with one-item additive loss.
Integer options	All options remain present, but their structural role is opaque.	Upper hulls are LP-safe only; exact search preserves below-hull non-dominated options.
Limited runs	Solver gap semantics.	Fixed- θ LP bounds, robust-feasible incumbents, and per- θ initialized global upper-bound accounting.
Failure diagnosis	Generic timeout or branch-and-bound log.	Tight capacity, weak θ pruning, and node-bound bottlenecks are visible as separate certificate objects.

The contribution is an auditable decomposition that identifies which part of the robust-MCKP structure controls feasibility, rounding loss, and failure to close.

3 Robust Pricing under Demand Uncertainty

In the formulation presented in [Section 1](#), the demand response $\hat{g}_i(x_i)$ is deterministic. To model demand-estimation error, we use budgeted robustness: the realized demand $\tilde{g}_i(x_i)$ lies in an uncertainty set $\mathcal{U}$, and the margin constraint must hold for all realizations in that set. The decision problem is to maximize nominal revenue subject to this worst-case feasibility requirement.

3.1 Uncertainty Model

We assume the realized demand lies within a symmetric interval around the nominal value:

$$\tilde{g}_i(x_i) \in [\hat{g}_i(x_i) - \delta_i(x_i), \hat{g}_i(x_i) + \delta_i(x_i)], \quad (11)$$

where $\delta_i(x_i) \geq 0$ is a variability parameter (e.g., derived from the standard error of the demand estimation). We assume $\delta_i(x_i) \leq \hat{g}_i(x_i)$ on X_i^{ad} , so demand remains nonnegative under uncertainty. For the robust ratio interpretation, we additionally assume that

$$\tilde{D}(\mathbf{x}) := \sum_{i=1}^n \omega_i a_i \tilde{g}_i(x_i) > 0$$

for every admissible price vector and every realization in the uncertainty set. The condition $\delta_i(x_i) \leq \hat{g}_i(x_i)$ guarantees nonnegative realized demand, but it does not by itself imply a strictly positive denominator unless at least one positive-exposure, positive-reference-price item retains positive realized demand in every admissible realization. If this positive-volume condition is not imposed, all robust results below should be read as applying to the additive residual constraint $\tilde{N}(\mathbf{x}) - \Delta \tilde{D}(\mathbf{x}) \geq 0$, which is the algebraic primitive used by the algorithms.

To avoid overly conservative solutions—such as those arising from box uncertainty where every product realizes its worst-case simultaneously—we adopt the “Budget of Uncertainty” approach proposed by [Bertsimas and Sim \(2004\)](#). We focus on integer cardinality budgets and define $\Gamma \in \{0, 1, \dots, n\}$ as the maximum number of products that can simultaneously deviate to their worst-case limits.

Definition 1 (Uncertainty Set). Let $\xi_i \in [-1, 1]$ be the scaled deviation for item i . For a fixed price vector $\mathbf{x}$, we define the uncertainty set $\mathcal{U}^\Gamma$ as:

$$\mathcal{U}^\Gamma(\mathbf{x}) = \left\{ \tilde{\mathbf{g}}(\mathbf{x}) \in \mathbb{R}^n \mid \tilde{g}_i(x_i) = \hat{g}_i(x_i) + \xi_i \delta_i(x_i), \quad \sum_{i=1}^n |\xi_i| \leq \Gamma, \quad |\xi_i| \leq 1 \quad \forall i \right\}. \quad (12)$$

3.2 The Robust Counterpart

Under the positive-denominator assumption above, the robust ratio constraint $\tilde{N}(\mathbf{x})/\tilde{D}(\mathbf{x}) \geq \Delta$ for all $\tilde{\mathbf{g}} \in \mathcal{U}^\Gamma(\mathbf{x})$ is equivalent to requiring the residual $\tilde{N}(\mathbf{x}) - \Delta \tilde{D}(\mathbf{x}) \geq 0$ for all realizations. Without that assumption, we take the residual inequality itself as the primitive robust feasibility condition. Substituting the definitions of $\tilde{N}(\mathbf{x})$ and $\tilde{D}(\mathbf{x})$, this condition is equivalent to:

$$\min_{\tilde{\mathbf{g}} \in \mathcal{U}^\Gamma(\mathbf{x})} \sum_{i=1}^n \omega_i \tilde{g}_i(x_i) (x_i - \Delta a_i) \geq 0. \quad (13)$$

Let $s_i(x_i) = \omega_i (x_i - \Delta a_i) \hat{g}_i(x_i)$ and $t_i(x_i) := \omega_i (x_i - \Delta a_i) \delta_i(x_i)$. The robust constraint can be rewritten as:

$$\sum_{i=1}^n s_i(x_i) + \min_{\xi: \sum |\xi_i| \leq \Gamma, |\xi_i| \leq 1} \sum_{i=1}^n t_i(x_i) \xi_i \geq 0. \quad (14)$$

The minimization term represents the worst-case deviation permitted by the budget. Since the objective is linear in ξ_i , the minimum is achieved when ξ_i takes values on the boundary of its domain to oppose the sign of $\omega_i(x_i - \Delta a_i)$. This worst-case behavior leads to the following exact formulation:

Theorem 2 (Robust Counterpart Formulation). *Let $\Gamma \in \{0, 1, \dots, n\}$ be an integer. Define*

$$\beta(\mathbf{x}, \Gamma) := \sum_{k=1}^{\Gamma} |t|_{(k)}(\mathbf{x}),$$

where $|t|_{(1)}(\mathbf{x}) \geq \dots \geq |t|_{(n)}(\mathbf{x})$ denotes the nonincreasing ordering of the numbers $\{|t_i(x_i)|\}_{i=1}^n$. We adopt the convention that $\beta(\mathbf{x}, 0) = 0$. Then the robust optimization problem is equivalent to

$$\max_{\mathbf{x}} \quad \sum_{i=1}^n \omega_i x_i \hat{g}_i(x_i) \tag{15}$$

$$s.t. \quad \sum_{i=1}^n s_i(x_i) - \beta(\mathbf{x}, \Gamma) \geq 0, \tag{16}$$

$$x_i \in [(1 - \sigma_i)a_i, (1 + \sigma_i)a_i] \cap \mathcal{X}_i \quad (\forall i).$$

Proof. Fix $\mathbf{x}$. By symmetry, an optimal deviation chooses the sign opposite to $t_i(x_i)$ whenever item i is used in the budget. With $y_i := |\xi_i|$, the inner minimization becomes

$$-\max \left\{ \sum_{i=1}^n |t_i(x_i)| y_i : \sum_i y_i \leq \Gamma, 0 \leq y_i \leq 1 \right\}.$$

For integer Γ , this linear program has an optimal extreme point that sets $y_i = 1$ on the Γ largest coefficients $|t_i(x_i)|$ and zero otherwise. The worst-case term is therefore $-\beta(\mathbf{x}, \Gamma)$, and substituting it into (14) gives (16). $\square$

3.3 Integer-budget scope and fractional-budget extension

Throughout the paper, Γ is an integer in $\{0, \dots, n\}$. Noninteger budgets in the Bertsimas–Sim model are standard, but then the worst-case penalty includes a fractional contribution from the next-largest coefficient rather than the simple sum of the Γ largest values. This is not a conceptual obstacle to the scalar θ representation, which remains the natural starting point for fractional budgets. It is, however, outside the validated code path here: the direct sorted-sum certificate, exact solver status logic, and regression tests are implemented for integer budgets only. Fractional-budget certification is therefore left to a separate implementation and validation pass.

4 Structural Analysis and Greedy Optimality

The Γ -budget penalty $\beta(\mathbf{x}, \Gamma)$ couples items and breaks separability, which is a main source of computational difficulty in robust discrete pricing. Even without robustness, the reduced problem is NP-hard because it is an MCKP. The key structural observation is that the LP relaxation remains efficiently solvable: each item’s discrete menu induces an upper hull in the cost–value plane, and the global LP reduces to greedy filling over hull segments. The sequel derives this hull-greedy LP structure and connects it to the robust model through the parametric decomposition used in Section 5.

4.1 Upper-hull geometry of the LP relaxation

Fix an item i and consider its admissible discrete options $x_{i,j} \in \mathcal{X}_i^{\text{ad}}$ from [Section 1](#). Each option induces a point (c_{ij}, v_{ij}) in the cost–value plane, and we write

$$\mathcal{P}_i := \{(c_{ij}, v_{ij}) : x_{i,j} \in \mathcal{X}_i^{\text{ad}}\} \subset \mathbb{R}^2.$$

In the LP relaxation, the constraints $\sum_j z_{ij} = 1$ and $z_{ij} \geq 0$ allow fractional mixing across options; equivalently, item i may realize any aggregate point $(c_i, v_i) \in \text{conv}(\mathcal{P}_i)$. Since the objective maximizes value for a given resource usage, only points on the *upper hull* of $\text{conv}(\mathcal{P}_i)$ can be optimal. This motivates *upper-hull filtering*: we replace $\mathcal{P}_i$ by the ordered vertex sequence $\mathcal{H}_i$ of the upper hull of $\text{conv}(\mathcal{P}_i)$. This removes LP-irrelevant points and does not change the LP relaxation optimum; as exact discrete preprocessing, however, only Pareto-dominated points can be removed. Points below the upper hull may still be relevant to the integer MCKP, so upper-hull filtering is used for the LP benchmark and rounding analysis, not as a claim of exact IP preprocessing.

Remark 3 (Upper hulls are not integer preprocessing). Consider a single MCKP group with options $(c, v) = (0, 0), (1, 1), (2, 3)$ and capacity $C = 1$. The middle option lies below the upper-hull segment joining $(0, 0)$ and $(2, 3)$, so it is unnecessary for the LP upper envelope. It is nevertheless the unique optimal integer choice: the option $(2, 3)$ is infeasible and dropping $(1, 1)$ would reduce the integer optimum from 1 to 0. Thus upper-hull filtering is valid for LP bounds, but not for exact integer preprocessing unless additional dominance conditions hold.

The theorem below states the LP structure, including the infeasible case when capacity is below the sum of the least-cost efficient hull vertices.

Remark 4 (Relation to prior work). The properties collected in [Theorem 5](#) are individually standard in the MCKP literature; see [Dyer \(1984\)](#), [Pisinger \(1995\)](#), and [Kellerer et al. \(2004, Chapter 11\)](#). We re-prove them in unified form because our notation differs from these sources and the segment parameterization is used directly in [Section 5](#). Readers familiar with the classical MCKP LP structure may skip ahead to the scalar θ decomposition without loss of continuity.

Theorem 5 (Hull reduction and greedy optimality for the LP relaxation). *Consider the LP relaxation of the Multiple-Choice Knapsack Problem (MCKP):*

$$\begin{aligned} \max_z \quad & \sum_{i=1}^n \sum_{j=1}^m v_{ij} z_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^n \sum_{j=1}^m c_{ij} z_{ij} \leq C, \\ & \sum_{j=1}^m z_{ij} = 1 \quad (i = 1, \dots, n), \\ & z_{ij} \geq 0 \quad (\forall i, j). \end{aligned} \tag{17}$$

For each item i , define the point set $\mathcal{P}_i := \{(c_{ij}, v_{ij}) : j = 1, \dots, m\} \subset \mathbb{R}^2$. First merge equal-cost points by keeping the largest value and remove Pareto-dominated points. The efficient upper-hull vertices can then be ordered as

$$\mathcal{H}_i = \{(c_{i,1}^H, v_{i,1}^H), \dots, (c_{i,p_i}^H, v_{i,p_i}^H)\}, \quad c_{i,1}^H < \dots < c_{i,p_i}^H, \quad v_{i,1}^H < \dots < v_{i,p_i}^H.$$

For $k = 1, \dots, p_i - 1$ define the hull segment lengths and slopes

$$\Delta c_{i,k}^H := c_{i,k+1}^H - c_{i,k}^H > 0, \quad \rho_{i,k}^H := \frac{v_{i,k+1}^H - v_{i,k}^H}{c_{i,k+1}^H - c_{i,k}^H}.$$

If $p_i = 1$, item i has no segment variables. If

$$C < \sum_{i=1}^n c_{i,1}^H,$$

then the LP relaxation is infeasible. Assume henceforth that $C \geq \sum_{i=1}^n c_{i,1}^H$.

Then the following statements hold.

- (i) (**Upper-hull sufficiency / two-adjacent-vertices mixing.**) There exists an optimal solution to (17) such that for each i , the induced aggregate point $(c_i, v_i) := (\sum_j c_{ij} z_{ij}, \sum_j v_{ij} z_{ij})$ lies on the upper hull of $\text{conv}(\mathcal{P}_i)$, and hence can be represented as a convex combination of at most two adjacent hull vertices $(c_{i,k}^H, v_{i,k}^H)$ and $(c_{i,k+1}^H, v_{i,k+1}^H)$ for some k .
- (ii) (**Reduction to continuous knapsack over hull segments.**) Under this feasibility condition, the exact upper-hull segment representation is

$$\begin{aligned} \max_{\ell} \quad & \sum_{i=1}^n v_{i,1}^H + \sum_{i=1}^n \sum_{k=1}^{p_i-1} \rho_{i,k}^H \ell_{i,k} \\ \text{s.t.} \quad & \sum_{i=1}^n \sum_{k=1}^{p_i-1} \ell_{i,k} \leq C - \sum_{i=1}^n c_{i,1}^H, \\ & 0 \leq \ell_{i,k} \leq \Delta c_{i,k}^H \quad (\forall i, k), \\ & \ell_{i,k+1} > 0 \Rightarrow \ell_{i,k} = \Delta c_{i,k}^H \quad (\forall i, k = 1, \dots, p_i - 2), \end{aligned} \tag{18}$$

where $\ell_{i,k}$ denotes the amount of capacity allocated to hull segment k of item i . Because efficient upper-hull slopes are nonincreasing within each item, dropping the prefix constraints does not change the optimal value: there exists an optimal solution of the relaxed segment problem satisfying them. Thus

$$\text{OPT}_{\text{LP}} = \sum_i v_{i,1}^H + \text{OPT}_{\text{seg}},$$

where

$$\text{OPT}_{\text{seg}} := \max_{\ell} \left\{ \sum_{i,k} \rho_{i,k}^H \ell_{i,k} : \sum_{i,k} \ell_{i,k} \leq C - \sum_i c_{i,1}^H, 0 \leq \ell_{i,k} \leq \Delta c_{i,k}^H \right\}.$$

- (iii) (**Greedy optimality for the LP relaxation.**) Let ℓ^* be obtained by filling positive-density efficient hull segments in nonincreasing order of densities $\rho_{i,k}^H$, each up to its upper bound $\Delta c_{i,k}^H$, while capacity remains and the next density is positive. Break ties within each item from earlier to later segments, so the selected segment vector can be chosen prefix-feasible. If several boundary segments have the same density, choose an extreme point of the tied optimal face, saturating all but possibly one of them. Then ℓ^* attains OPT_{seg} and induces an optimal solution to the LP relaxation (17). In particular, even under equal segment densities, there exists an optimal LP solution in which at most one item is mixed fractionally (i.e. across two adjacent hull vertices); all other items select a single hull vertex.
- (iv) (**Special case: incremental greedy on the raw chain.**) Suppose for every i the raw point chain $\{(c_{i,j}, v_{i,j})\}_{j=1}^m$ satisfies $c_{i,1} < \dots < c_{i,m}$ and, after dominance cleanup, coincides with the efficient upper-hull vertex sequence. Then the greedy rule in (iii) reduces to the incremental greedy algorithm on adjacent increments $\rho_{i,j} = (v_{i,j} - v_{i,j-1}) / (c_{i,j} - c_{i,j-1})$.

Proof. The convex-hull reformulation, restriction to the upper hull, and segment-length parameterization give statements (i)–(ii). These are standard arguments adapted to the notation of this paper; the details are provided in [Appendix A](#) for completeness.

Proof of (iii). After the constant base value $\sum_i v_{i,1}^H$ is accounted for, the relaxed segment problem defining OPT_{seg} is a continuous knapsack problem with items indexed by pairs (i, k) , value density $\rho_{i,k}^H$, and capacity upper bound $\Delta c_{i,k}^H$. A standard exchange argument shows that an optimal solution is obtained by filling variables in nonincreasing order of $\rho_{i,k}^H$: take two variables (i, k) and (i', k') with $\rho_{i,k}^H > \rho_{i',k'}^H$. If $0 < \ell_{i',k'} < \Delta c_{i',k'}^H$ while $\ell_{i,k} < \Delta c_{i,k}^H$, then shifting an infinitesimal amount ε of capacity from $\ell_{i',k'}$ to $\ell_{i,k}$ increases the objective by $\varepsilon(\rho_{i,k}^H - \rho_{i',k'}^H) > 0$ while preserving feasibility, contradicting optimality. Thus, in any optimum, all higher-density variables are saturated before any lower-density variable is positive, implying the greedy fill rule is optimal for the continuous segment-knapsack value. When the capacity cutoff falls inside a set of equal-density variables, the optimal solutions over that tied face form a polytope obtained by one residual capacity equality and bound constraints; an extreme point of this face has at most one tied variable strictly between its bounds. Selecting such an extreme point gives an optimal segment solution with at most one fractional segment. The prefix-restoration argument in [Appendix A](#), together with the earlier-before-later tie-breaking within each item, ensures that the greedy segment vector can be chosen to induce a valid upper-hull LP solution without lowering value. This proves (iii).

Proof of (iv). If for each i the dominance-cleaned raw chain coincides with the efficient upper-hull vertices, i.e. $\mathcal{H}_i = \{(c_{i,j}, v_{i,j})\}_{j=1}^m$ after cleanup, then the hull segments are exactly the adjacent increments $(j-1) \rightarrow j$ and $\rho_{i,k}^H$ equals $\rho_{i,j}$. Therefore the greedy rule in (iii) reduces to selecting adjacent upgrades in descending order of $\rho_{i,j}$, which is precisely the incremental greedy algorithm. $\square$

4.2 Robust constraint structure: scalar θ decomposition

For the coupled integer-budget case $\Gamma < n$, items interact through the shared uncertainty budget. By [Theorem 2](#), the robust margin constraint is equivalent to

$$\sum_{i=1}^n s_i(x_i) - \beta(\mathbf{x}, \Gamma) \geq 0, \quad (19)$$

where $\beta(\mathbf{x}, \Gamma) = \sum_{k=1}^{\Gamma} |t|_{(k)}(\mathbf{x})$ is the sum of the Γ largest values of $\{|t_i(x_i)|\}_{i=1}^n$. While this characterization is explicit, direct optimization over $\beta(\mathbf{x}, \Gamma)$ is difficult because it couples all items through a sorting operation. We resolve this by observing that $\beta(\mathbf{x}, \Gamma)$ admits a dual scalar representation:

$$\beta(\mathbf{x}, \Gamma) = \max_{\xi} \left\{ \sum_{i=1}^n |t_i(x_i)| \xi_i \mid \sum_{i=1}^n \xi_i \leq \Gamma, 0 \leq \xi_i \leq 1 \forall i \right\}. \quad (20)$$

Theorem 6 (Parametric decomposition of the coupled Γ -budget penalty). *Let $\Gamma \in \{0, \dots, n\}$ be an integer. For every $\mathbf{x}$, the penalty $\beta(\mathbf{x}, \Gamma)$ in (20) admits the dual representation*

$$\beta(\mathbf{x}, \Gamma) = \min_{\theta \geq 0} \left\{ \Gamma \theta + \sum_{i=1}^n \max \{0, |t_i(x_i)| - \theta\} \right\}. \quad (21)$$

Consequently, the robust margin constraint (19) holds if and only if there exists $\theta \geq 0$ such that

$$\sum_{i=1}^n \left(s_i(x_i) - \max \{0, |t_i(x_i)| - \theta\} \right) \geq \Gamma \theta. \quad (22)$$

For any fixed $\theta \geq 0$, the left-hand side of (22) is separable across items.

Proof. Fix $\mathbf{x}$. Problem (20) is a feasible and bounded linear program over a nonempty compact polytope. Its dual is

$$\min_{\theta \geq 0, \pi \geq 0} \Gamma\theta + \sum_{i=1}^n \pi_i \quad \text{s.t.} \quad \theta + \pi_i \geq |t_i(x_i)| \quad (\forall i).$$

Strong duality yields equality of primal and dual optimal values. For any fixed $\theta \geq 0$, the minimal feasible choice is $\pi_i = \max\{0, |t_i(x_i)| - \theta\}$, which gives (21). Substituting (21) into (19) yields (22). $\square$

5 A Robust Hull-Greedy Algorithm for the Discrete Problem

We develop a method for the *discrete* Γ -budget robust pricing problem. After the reduction in Section 1, the problem is an MCKP and remains NP-hard, so we target exact robust-feasibility certification together with explicit performance guarantees relative to LP benchmarks. The algorithm is driven by two structural results: the one-dimensional parametric reformulation of the coupled robust constraint (Theorem 6) and the exact hull-greedy solution of the fixed-parameter LP relaxation (Theorem 5).

5.1 Exact robust feasibility via a scalar parameter

For each admissible option $x_{i,j} \in \mathcal{X}_i^{\text{ad}}$, define

$$v_{ij} := \omega_i x_{i,j} \hat{g}_i(x_{i,j}), \quad s_{ij} := \omega_i (x_{i,j} - \Delta a_i) \hat{g}_i(x_{i,j}), \quad t_{ij} := \omega_i (x_{i,j} - \Delta a_i) \delta_i(x_{i,j}).$$

By Theorem 6, the robust margin constraint is equivalent to the existence of $\theta \geq 0$ such that

$$\sum_{i=1}^n \left(s_i(x_i) - \max\{0, |t_i(x_i)| - \theta\} \right) \geq \Gamma\theta. \quad (23)$$

For fixed $\theta \geq 0$, introduce the per-option θ -modified contributions

$$s_{ij}^\theta := s_{ij} - \max\{0, |t_{ij}| - \theta\}. \quad (24)$$

Then (23) becomes the separable inequality

$$\sum_{i=1}^n s_{i,j(i)}^\theta \geq \Gamma\theta, \quad j(i) \in \{1, \dots, m\} \text{ chosen for each } i. \quad (25)$$

where $j(i)$ denotes the index of the option selected for item i .

Proposition 7 (Finite full-breakpoint θ reduction). *Assume finite admissible menus and an integer uncertainty budget $\Gamma \in \{0, \dots, n\}$. Let*

$$\mathcal{B} := \{0\} \cup \{|t_{ij}| : i = 1, \dots, n, x_{i,j} \in \mathcal{X}_i^{\text{ad}}\}.$$

The union is over all original admissible options, before upper-hull filtering or dominance reductions used in LP-bound computations. If the discrete robust problem is feasible, then there exists an optimal solution $(\mathbf{x}^, \theta^*)$ with $\theta^* \in \mathcal{B}$. Equivalently, the robust optimum is the maximum of the fixed- θ MCKP optima over this finite set, with infeasible fixed- θ subproblems ignored. This is a mathematical reduction, not an implementation shortcut.*

Proof. Fix a discrete $\mathbf{x}$. Define

$$f_{\mathbf{x}}(\theta) := \sum_{i=1}^n \left(s_i(x_i) - \max\{0, |t_i(x_i)| - \theta\} \right) - \Gamma\theta.$$

Then $f_{\mathbf{x}}$ is concave piecewise-affine in θ , with breakpoints contained in $\{0\} \cup \{|t_i(x_i)|\}_{i=1}^n$. If $f_{\mathbf{x}}(\theta_0) \geq 0$ for some $\theta_0 \geq 0$, let the maximal affine interval containing θ_0 be $[\theta_L, \theta_U]$, allowing $\theta_U = \infty$. If $\theta_U < \infty$, then $f_{\mathbf{x}}(\theta_0)$ is a convex combination of the two endpoint values, so at least one endpoint satisfies $f_{\mathbf{x}}(\theta) \geq 0$. If the interval is unbounded above, then its finite left endpoint is a breakpoint; since $f_{\mathbf{x}}$ is affine on that interval and has slope $-\Gamma$ after all $|t_i(x_i)|$ are below θ (or slope 0 when $\Gamma = 0$), feasibility at an interior point implies feasibility at the finite endpoint. Thus feasibility for fixed $\mathbf{x}$ can always be certified by some breakpoint $\theta \in \{0\} \cup \{|t_i(x_i)|\}_{i=1}^n$. The case $\Gamma = 0$ is included: once θ reaches the largest selected uncertainty magnitude, $f_{\mathbf{x}}$ is constant, and feasibility on the unbounded terminal interval is certified at its left endpoint. Because each $|t_i(x_i)|$ equals $|t_{i,j(i)}|$ for some original admissible option, every such value belongs to $\mathcal{B}$. The nominal objective $\sum_i v_{i,j(i)}$ is independent of θ , so replacing any feasible certificate θ_0 for a fixed selection by a feasible breakpoint does not change that selection's objective value. Since this argument holds for every discrete $\mathbf{x}$ individually, exact robust optimization is exactly the maximum over the fixed- θ discrete MCKPs for $\theta \in \mathcal{B}$. Algorithm 1 instead solves the fixed- θ LP relaxation exactly and applies a feasibility-preserving rounding step with the additive guarantees below. $\square$

Remark 8 (Reduced breakpoint sets can be unsafe). The full original candidate set in Proposition 7 is essential. For illustration, take $n = 2$, $\Gamma = 1$, and a fixed selection with $s_1 = s_2 = 3/4$ and $|t_1| = |t_2| = 1$. Its direct robust residual is $s_1 + s_2 - \max\{|t_1|, |t_2|\} = 1/2$, so the selection is robust feasible. The corresponding certificate function is

$$f(\theta) = \frac{3}{2} - 2 \max\{0, 1 - \theta\} - \theta.$$

At $\theta = 0$, $f(\theta) = -1/2$, while at the original breakpoint $\theta = 1$, $f(\theta) = 1/2$. Omitting that breakpoint can therefore falsely reject a feasible robust selection. The example is only a warning about reduced enumeration; it is not a complexity statement.

5.2 Fixed- θ reformulation as an MCKP

Fix $\theta \in \mathcal{B}$. Convert (25) into a knapsack inequality with nonnegative costs via a baseline-slack transformation. For each item i , choose

$$j^\theta(i) \in \arg \max_j s_{ij}^\theta, \quad S_{\max}^\theta := \sum_{i=1}^n s_{i,j^\theta(i)}^\theta, \quad c_{ij}^\theta := s_{i,j^\theta(i)}^\theta - s_{ij}^\theta \ (\geq 0).$$

Then (25) is equivalent to

$$\sum_{i=1}^n c_{i,j(i)}^\theta \leq C^\theta, \quad C^\theta := S_{\max}^\theta - \Gamma\theta. \quad (26)$$

Thus, for fixed θ , robust-feasible discrete pricing is exactly the MCKP

$$\max \sum_{i=1}^n v_{i,j(i)} \quad \text{s.t.} \quad \sum_{i=1}^n c_{i,j(i)}^\theta \leq C^\theta, \quad x_{i,j(i)} \in \mathcal{X}_i^{\text{ad}} \ (\forall i). \quad (27)$$

5.3 Upper-hull filtering and an exact LP solver for fixed θ

Fix $\theta \in \mathcal{B}$ with $C^\theta \geq 0$. Item i induces the point set

$$\mathcal{P}_i^\theta := \{(c_{ij}^\theta, v_{ij}) : x_{i,j} \in \mathcal{X}_i^{\text{ad}}\} \subset \mathbb{R}^2.$$

Before computing the hull, apply a standard preprocessing step: (a) if multiple points share the same cost c , keep only the one with largest value v ; (b) remove dominated points (if $c \leq c'$ and $v \geq v'$, discard (c', v')). This does not change the upper hull. After this preprocessing, the retained efficient hull vertices have strictly increasing costs and values, except in the degenerate case of a single retained vertex. Upper-hull filtering is used to solve the LP relaxation and to define the rounding benchmark. Only Pareto-dominated or equal-cost lower-value points are removed as integer-safe preprocessing; non-dominated below-hull options remain part of exact integer search. Apply *upper-hull filtering* by replacing $\mathcal{P}_i^\theta$ with the ordered vertex sequence

$$\mathcal{H}_i^\theta = \{(c_{i,1}^{H,\theta}, v_{i,1}^{H,\theta}), \dots, (c_{i,p_i}^{H,\theta}, v_{i,p_i}^{H,\theta})\}$$

of the upper hull of $\text{conv}(\mathcal{P}_i^\theta)$ (ordered by increasing c). Since $\mathcal{H}_i^\theta$ consists of vertices of $\text{conv}(\mathcal{P}_i^\theta)$, each hull vertex is one of the original menu points (c_{ij}^θ, v_{ij}) and therefore corresponds to a discrete option. Define hull segment lengths and slopes

$$\Delta c_{i,k}^{H,\theta} := c_{i,k+1}^{H,\theta} - c_{i,k}^{H,\theta} > 0, \quad \rho_{i,k}^{H,\theta} := \frac{v_{i,k+1}^{H,\theta} - v_{i,k}^{H,\theta}}{\Delta c_{i,k}^{H,\theta}} \quad (k = 1, \dots, p_i - 1).$$

By [Theorem 5](#), the LP relaxation of (27) is solved *exactly* by greedily filling hull segments in nonincreasing order of $\rho_{i,k}^{H,\theta}$. Moreover, there exists an optimal LP solution in which at most one item is fractional (mixed between two adjacent hull vertices).

Proposition 9 (Fixed- θ roles of hull geometry). *For a fixed θ with $C^\theta \geq 0$, the upper-hull construction has three distinct roles. First, it gives the exact LP relaxation value by the greedy segment oracle of [Theorem 5](#). Second, the corresponding LP solution can be chosen with at most one fractionally mixed class, which yields the one-item HullRound certificate in [Propositions 10](#) and [11](#). Third, in exact branch-and-bound the same hull value is a valid upper bound at nodes. None of these roles permits discarding non-dominated below-hull integer options from the exact fixed- θ MCKP; exactness requires branching over the original non-dominated options.*

Proof. The first two statements are direct consequences of [Theorem 5](#): the LP optimum is achieved on the upper hull and has an extreme optimal solution with at most one fractional segment. Rounding the single mixed class down preserves fixed- θ feasibility and loses at most the adjacent item-value jump. For a branch-and-bound node, every integer completion is feasible for the residual LP relaxation, so the hull-greedy LP value is an upper bound. [Remark 3](#) shows why this LP irrelevance is not an integer-preprocessing certificate; only ordinary dominance can remove integer options without changing the fixed- θ discrete problem. $\square$

Complexity. Let $M = \sum_i |J_i^{\text{ad}}|$ be the total number of admissible menu options and let $H = \sum_i (p_i - 1)$ be the total number of fixed- θ upper-hull segments. For a fixed θ , constructing the item hulls costs roughly $O(\sum_i |J_i^{\text{ad}}| \log |J_i^{\text{ad}}|)$ unless transformed point sets are cached. The fixed- θ LP evaluation then sorts or scans the H hull segments, with cost depending on whether the segment order is rebuilt or reused. Independent enumeration multiplies the fixed- θ work by $|\mathcal{B}|$, while exact branch-and-bound remains exponential in the worst case. Incremental θ sweeps can reduce repeated work in implementations, but the exact sweep is not a worst-case complexity improvement claim.

Exact robust-feasibility certificate (final check). For any discrete price vector $\mathbf{x}$, robust feasibility can be certified directly by

$$\sum_{i=1}^n s_i(x_i) - \beta(\mathbf{x}, \Gamma) \geq 0, \quad \beta(\mathbf{x}, \Gamma) = \sum_{k=1}^{\Gamma} |t|_{(k)}(\mathbf{x}), \quad (28)$$

where $t_i(\mathbf{x}) = \omega_i(\mathbf{x} - \Delta a_i)\delta_i(\mathbf{x})$ and $|t|_{(1)}(\mathbf{x}) \geq \dots \geq |t|_{(n)}(\mathbf{x})$ are the sorted magnitudes.

5.4 Discrete recovery: one-item rounding with optional repair

Let $\mathbf{z}^{\text{LP}}(\theta)$ be an optimal LP solution produced by the hull-greedy procedure in [Section 5.3](#). If no item is fractional, it is already discrete. Otherwise, exactly one item i^* is mixed between two adjacent hull vertices $(c_{i^*,k}^{H,\theta}, v_{i^*,k}^{H,\theta})$ and $(c_{i^*,k+1}^{H,\theta}, v_{i^*,k+1}^{H,\theta})$.

Default choice (feasibility-preserving round-down). Rounding i^* *down* to the adjacent vertex with smaller knapsack cost c^θ weakly decreases total cost and thus preserves (26); all other items remain unchanged. This yields a feasible discrete solution $\mathbf{x}^\downarrow(\theta)$.

Optional improvement (round-up + repair). In practice, rounding down can lose a larger single-item value jump than necessary. As an optional refinement, also consider rounding i^* *up* to the higher vertex, obtaining $\mathbf{x}^\uparrow(\theta)$, which may violate (26) by some excess cost $\Delta C > 0$. If $\Delta C > 0$, attempt a *repair* step by performing discrete *downgrades* along hull chains: repeatedly select a downgrade that reduces knapsack cost while minimizing loss of objective, specifically by selecting the downgrade $(i, k \rightarrow k-1)$ that minimizes the ratio $(v_{i,k}^{H,\theta} - v_{i,k-1}^{H,\theta}) / (c_{i,k}^{H,\theta} - c_{i,k-1}^{H,\theta})$ among items with remaining downgrade capacity. If repair succeeds, keep the repaired candidate; if no sequence of allowed downgrades restores feasibility, discard the round-up candidate. Alternatively, allowing the rounded-up item itself to be downgraded as a last resort returns to the round-down candidate, so the overall procedure is always feasibility-preserving because the round-down candidate is always available. Finally, keep the best feasible candidate,

$$\mathbf{x}(\theta) \in \arg \max \{N(\mathbf{y}) : \mathbf{y} \in \mathcal{F}(\theta)\},$$

where $\mathcal{F}(\theta)$ contains $\mathbf{x}^\downarrow(\theta)$ and any feasible repaired round-up candidate retained by the repair step. This refinement is heuristic but feasibility-preserving. All theoretical guarantees apply to the round-down solution $\mathbf{x}^\downarrow(\theta)$ alone; see [Propositions 10](#) and [11](#). The optional round-up-and-repair and completion steps can only improve the objective and are not required for the stated bounds; they are not used to assert fixed- θ or global optimality.

Feasibility-preserving completion. While capacity remains, apply discrete *upgrades* along each item's hull chain. Consider only adjacent-vertex upgrades that fit and weakly improve the objective, and process them in decreasing order of slope. This preserves the fixed- θ knapsack constraint.

5.5 Complete algorithm and guarantees

We enumerate $\theta \in \mathcal{B}$ (which is sufficient by [Proposition 7](#)), solve the corresponding LP relaxation exactly by hull-greedy, and recover a discrete feasible solution by one-item rounding (optionally with repair/completion); see [Algorithm 1](#).

Proposition 10 (Additive performance bound (per fixed θ)). *Fix $\theta \in \mathcal{B}$ with $C^\theta \geq 0$. Let $\text{OPT}_{\text{LP}}(\theta)$ and $\text{OPT}_{\text{IP}}(\theta)$ denote the optimal objective values of the LP relaxation of (27) and of (27), respectively. Let*

$\mathbf{x}^\downarrow(\theta)$ be the round-down solution. Then

$$\text{OPT}_{\text{LP}}(\theta) - N(\mathbf{x}^\downarrow(\theta)) \leq \Delta V_{\max}^\theta, \quad \text{and hence} \quad \text{OPT}_{\text{IP}}(\theta) - N(\mathbf{x}^\downarrow(\theta)) \leq \Delta V_{\max}^\theta,$$

where $\Delta V_{\max}^\theta$ is defined in [Section 5.6](#). Any optional repair/completion candidate retained by [Algorithm 1](#) is feasible and has objective at least as large as the round-down candidate selected for that θ , so the same bound applies a fortiori to the final retained candidate whenever it improves on round-down.

Proof. By [Theorem 5\(iii\)](#), there exists an optimal LP solution mixing at most one item between two adjacent hull vertices. Rounding down that item (to the adjacent vertex with smaller c^θ) preserves feasibility and decreases the objective by at most the corresponding adjacent value jump, bounded by $\Delta V_{\max}^\theta$. The second inequality follows from $\text{OPT}_{\text{LP}}(\theta) \geq \text{OPT}_{\text{IP}}(\theta)$. Optional repair and completion retain only feasible candidates whose objective weakly improves on $\mathbf{x}^\downarrow(\theta)$, so replacing the round-down candidate by such a candidate cannot weaken the bound. $\square$

Global implication across θ . Let

$$\mathcal{B}_{\text{feas}} := \{\theta \in \mathcal{B} : C^\theta \geq 0\}.$$

For $\theta \in \mathcal{B}_{\text{feas}}$, let $\mathbf{x}^\downarrow(\theta)$ denote the round-down solution produced from the fixed- θ LP relaxation, and define

$$R := \max_{\theta \in \mathcal{B}_{\text{feas}}} N(\mathbf{x}^\downarrow(\theta)).$$

This is the best round-down value over the evaluated candidates. Let

$$\text{OPT}_{\text{IP}}^{\text{rob}} := \max_{\theta \in \mathcal{B}_{\text{feas}}} \text{OPT}_{\text{IP}}(\theta)$$

be the exact robust discrete optimum, using [Proposition 7](#). Then

$$\text{OPT}_{\text{IP}}^{\text{rob}} - R \leq \max_{\theta \in \mathcal{B}_{\text{feas}}} \Delta V_{\max}^\theta.$$

Similarly,

$$\max_{\theta \in \mathcal{B}_{\text{feas}}} \text{OPT}_{\text{LP}}(\theta) - R \leq \max_{\theta \in \mathcal{B}_{\text{feas}}} \Delta V_{\max}^\theta.$$

If [Algorithm 1](#) keeps optional feasible improvements, its final objective is at least R , so the same bounds hold for the algorithm's final output.

Algorithm 1 Robust hull-greedy with θ -enumeration and discrete recovery

```

1: Input: admissible menus  $\{\mathcal{X}_i^{\text{ad}}\}$ , data  $(v_{ij}, s_{ij}, t_{ij})$ , budget  $\Gamma$ .
2: Form  $\mathcal{B} = \{0\} \cup \{|t_{ij}| : x_{i,j} \in \mathcal{X}_i^{\text{ad}}\}$ , deduplicate, and sort increasingly.
3:  $\text{BESTVAL} \leftarrow -\infty$ ,  $\text{BESTSOL} \leftarrow \emptyset$ .
4: for  $\theta \in \mathcal{B}$  do
5:   Compute  $s_{ij}^\theta$  via (24).
6:   Choose  $j^\theta(i) \in \arg \max_j s_{ij}^\theta$  and set  $c_{ij}^\theta = s_{i, j^\theta(i)}^\theta - s_{ij}^\theta$ .
7:    $C^\theta \leftarrow \sum_{i=1}^n s_{i, j^\theta(i)}^\theta - \Gamma\theta$ .
8:   if  $C^\theta < 0$  then continue.
9:   end if
10:  for  $i = 1, \dots, n$  do
11:    Build  $\mathcal{P}_i^\theta = \{(c_{ij}^\theta, v_{ij})\}_j$ .
12:    Preprocess  $\mathcal{P}_i^\theta$ : merge equal costs (keep max  $v$ ), then remove dominated points.
13:    Compute the upper hull  $\mathcal{H}_i^\theta$  and the slopes  $\rho_{i,k}^{H,\theta}$  and lengths  $\Delta c_{i,k}^{H,\theta}$ .
14:  end for
15:  Solve the LP relaxation of (27) exactly via hull-greedy over  $\rho_{i,k}^{H,\theta}$ .
16:  Recover a discrete solution  $\mathbf{x}(\theta)$  via Section 5.4 (round-down; optionally round-up+repair; optionally complete).
17:  Compute  $N(\mathbf{x}(\theta)) = \sum_{i=1}^n \omega_i x_i(\theta) \hat{g}_i(x_i(\theta))$ .
18:  Compute the exact robust certificate  $Z_\Gamma(\mathbf{x}(\theta)) := \sum_i s_i(x_i(\theta)) - \beta(\mathbf{x}(\theta), \Gamma)$  as in (28).
19:  if  $Z_\Gamma(\mathbf{x}(\theta)) < 0$  then continue.
20:  end if
21:  if  $N(\mathbf{x}(\theta)) > \text{BESTVAL}$  then
22:     $\text{BESTVAL} \leftarrow N(\mathbf{x}(\theta))$ ;  $\text{BESTSOL} \leftarrow (\theta, \mathbf{x}(\theta))$ .
23:  end if
24: end for
25: if  $\text{BESTSOL} = \emptyset$  then
26:   Output: No feasible solution found by the LP-rounding procedure.
27: else
28:   Output:  $\text{BESTSOL}$ .
29: end if

```

Practical note on θ enumeration. The set $\mathcal{B}$ can be as large as $|\mathcal{B}| \leq nm+1$. In heuristic or implementation-oriented variants, one may reduce the search by considering breakpoints induced by options that survive dominance or hull preprocessing; the exact guarantee, however, uses the full set $\mathcal{B}$. Moreover, (24) implies

$$s_{ij}^\theta = \begin{cases} s_{ij}, & \theta \geq |t_{ij}|, \\ s_{ij} - |t_{ij}| + \theta, & \theta < |t_{ij}|, \end{cases}$$

so s_{ij}^θ is piecewise-affine in θ with breakpoints at $|t_{ij}|$. The functions s_{ij}^θ have breakpoints at $|t_{ij}|$, but $\arg \max$ identities can also change at pairwise intersections of affine pieces. Exact enumeration in Algorithm 1 therefore uses the full set $\mathcal{B}$ and recomputes the per-item baseline at each enumerated θ . An exact sweep may reuse a previously computed hull only when the induced cost–value point set is identical; otherwise the hull is rebuilt. With this rule, the sweep constructs the same fixed- θ subproblems as independent enumeration. In floating-point implementations, equality checks use stated tolerances, and reported incumbents are checked against the original robust certificate (28).

Parametric sweep interpretation. For a fixed selection $\mathbf{x}$, the certificate $f_{\mathbf{x}}(\theta) = \sum_i (s_i(x_i) - (|t_i(x_i)| - \theta)_+) - \Gamma\theta$ is concave and piecewise affine. The implementation exploits the same structure by sweeping the full breakpoint set, incrementally updating s_{ij}^θ , recomputing baselines/costs/capacity, and reusing an item hull only when the induced cost–value point multiset and integer-safe option set are unchanged. This reuse

rule is exact-safe, but it is not an amortized speed theorem; the numerical section reports when maintenance overhead dominates.

5.6 One-item LP–IP gap certificate

For a fixed θ with $C^\theta \geq 0$, let $\mathcal{H}_i^\theta = \{(c_{i,1}^{H,\theta}, v_{i,1}^{H,\theta}), \dots, (c_{i,p_i}^{H,\theta}, v_{i,p_i}^{H,\theta})\}$ be the upper-hull vertex sequence for item i . Define

$$\Delta V_{\max}^\theta := \max_i \max_{k=1, \dots, p_i-1} (v_{i,k+1}^{H,\theta} - v_{i,k}^{H,\theta})_+,$$

with value zero when all hulls have one vertex. This is the largest objective loss incurred by rounding down a single fractional class.

Proposition 11 (Additive integrality-gap bound). *For every fixed $\theta \in \mathcal{B}$ with $C^\theta \geq 0$,*

$$0 \leq \text{OPT}_{\text{LP}}(\theta) - \text{OPT}_{\text{IP}}(\theta) \leq \Delta V_{\max}^\theta.$$

Proof. The LP relaxation contains the integer feasible set. By [Theorem 5](#), an optimal LP solution can be chosen with at most one fractional item, mixed between adjacent hull vertices. Rounding that item to the lower-cost endpoint preserves fixed- θ feasibility and loses at most the adjacent value jump, bounded by $\Delta V_{\max}^\theta$. The resulting integer point gives the stated upper bound. $\square$

The constant is the right scale: a one-item two-option instance with capacity just below the high-cost option can make the LP–IP gap approach the full adjacent value jump. In large nondegenerate portfolios this additive one-item loss becomes small relative to total value; the experiments report that relative behavior as a diagnostic rather than as an additional asymptotic claim. The diagnostic quantities are $L_{\text{rd}}/\Delta V_{\max}^\theta$, where L_{rd} is the realized LP-to-round-down loss, and $\Delta V_{\max}^\theta/\max\{|\text{UB}_{\text{LP}}(\theta)|, 1\}$ when the corresponding LP upper bound is recorded. These ratios measure certificate strength relative to the scale of the fixed- θ row; they do not replace the additive guarantee above. Optional repair and completion are reported as primal improvements only.

6 Exact Structure-Exploiting Branch-and-Bound

This section describes the exact certifying solver. It keeps the finite θ decomposition and hull-greedy LP structure from the preceding structural analysis and HullRound sections, but replaces LP rounding by exact fixed- θ branch-and-bound when an optimality certificate is required. HullRound remains useful as a primal heuristic and empirical baseline.

6.1 Motivation

The HullRound algorithm solves each fixed- θ LP relaxation exactly by upper-hull filtering and greedy filling over hull segments, then rounds the resulting solution to a robust-feasible discrete point. The finite-candidate result in [Proposition 7](#) shows that exact robust optimization over the original Γ -budget problem follows by solving the discrete MCKP (27) exactly for every $\theta \in \mathcal{B}$. One can implement this reduction by calling a generic MILP solver inside the θ loop. The proposed exact solver instead uses the hull-greedy LP relaxation as the bounding engine inside a specialized exact MCKP branch-and-bound.

The method is exact when all branch-and-bound nodes are fathomed and returns a valid anytime gap under time or node limits. Its purpose is not to replace mature MILP solvers, but to make the robust-MCKP decomposition algorithmically constructive: fixed- θ subproblems are solved by a structure-aware search, and the global robust solver combines those certificates with valid θ -level pruning.

6.2 Fixed- θ exact subproblem

For a fixed $\theta \in \mathcal{B}$ with $C^\theta \geq 0$, the exact subproblem is the MCKP

$$\begin{aligned} \max_{j(1), \dots, j(n)} \quad & \sum_{i=1}^n v_{i,j(i)} \\ \text{s.t.} \quad & \sum_{i=1}^n c_{i,j(i)}^\theta \leq C^\theta, \\ & j(i) \in J_i^{\text{ad}} \quad (i = 1, \dots, n). \end{aligned} \tag{29}$$

Equivalently, the decision selects exactly one option per item. The objective coefficients v_{ij} are independent of θ , while the costs c_{ij}^θ and capacity C^θ are induced by the robust-margin baseline transformation in [Section 5.2](#).

6.3 Node upper bounds

A branch-and-bound node is defined by a set of fixed assignments $F \subseteq \{1, \dots, n\}$ and selected options $j_F(i) \in J_i^{\text{ad}}$ for $i \in F$. The free item set is $U := \{1, \dots, n\} \setminus F$. The node stores the accumulated value and used capacity

$$V_F := \sum_{i \in F} v_{i,j_F(i)}, \quad C_F := \sum_{i \in F} c_{i,j_F(i)}^\theta,$$

and the remaining capacity

$$\widehat{C}_F^\theta := C^\theta - C_F.$$

If $\widehat{C}_F^\theta < 0$, the node is infeasible and is fathomed immediately. Otherwise, every completion of the node must choose one option for each item in U with total remaining cost at most $\widehat{C}_F^\theta$. The incumbent is the best feasible integer solution found so far for this fixed θ , with value denoted by IncVal_θ .

Proposition 12 (Hull-greedy node upper bound). *For any feasible branch-and-bound node F , the optimal value of the LP relaxation over the free items U , plus V_F , is an admissible upper bound on every integer completion of that node. In the implementation, this bound is computed by the upper-hull greedy method, without solving a generic LP or MILP at the node.*

Proof. Every integer completion of node F is feasible for the corresponding LP relaxation over the same free items and remaining capacity $\widehat{C}_F^\theta$. Therefore its remaining value is bounded above by the optimal free-item LP value. Adding the fixed contribution V_F gives a valid upper bound on the total completion value. Exact computability follows from applying [Theorem 5](#) to the residual fixed- θ MCKP induced by U and $\widehat{C}_F^\theta$. $\square$

6.4 Fixed- θ branch-and-bound

At each live node, the solver computes the hull-greedy LP relaxation over the free items. If the LP solution is integral, the node itself provides a feasible completion and the incumbent is updated if its value is better. If the LP solution is fractional, the standard hull-greedy structure gives an optimal LP solution with at most one item mixed between two adjacent hull vertices. The implementation branches on that fractional item when it is exposed; otherwise it uses a deterministic spread-based fallback. In both cases, the children cover the original fixed- θ integer options after only integer-safe dominance pruning.

This last point is essential. Upper-hull filtering is safe for the LP bound, but not as a general integer-preprocessing rule. The exact search therefore does not branch only over upper-hull vertices. It removes an option only when another option for the same item has weakly smaller fixed- θ cost and weakly larger value,

with at least one strict inequality; non-dominated below-hull options remain searchable and can be selected by the exact solver.

Theorem 13 (Fixed- θ branch-and-bound exactness). *Fix $\theta \in \mathcal{B}$ and suppose the fixed- θ branch-and-bound processes nodes until every node is fathomed. If branching covers every original non-dominated integer option for the selected item, and pruning is performed only when a node is infeasible or when its valid LP upper bound is no larger than the incumbent value, then the algorithm returns an optimal solution of the fixed- θ MCKP (29).*

Proof. The per-item dominance rule preserves at least one optimal completion because any removed option can be replaced within the same item by an option with no larger fixed- θ cost and no smaller value. The remaining branching scheme partitions the non-dominated integer completions of each unfathomed node into child nodes without discarding feasible assignments. Infeasible nodes contain no valid completion. By [Proposition 12](#), a node whose LP upper bound is no larger than the incumbent cannot contain a completion that improves the incumbent. When all nodes are fathomed, every feasible integer solution has either been evaluated through an integral node or belongs to a pruned node that cannot improve the incumbent. Hence the incumbent is optimal. $\square$

Implementation note. The theorem is an exact-arithmetic statement. In the floating-point experiments, pruning comparisons use stated numerical tolerances, and any reported incumbent is checked against the original robust certificate (28).

6.5 Global θ -enumerated exact solver

The fixed- θ solver is embedded in the finite candidate enumeration from [Proposition 7](#). The implementation constructs the full candidate set

$$\mathcal{B} = \{0\} \cup \{|t_{ij}| : i = 1, \dots, n, j \in J_i^{\text{ad}}\},$$

using all original admissible options. It does not use the reduced candidate sets sometimes sufficient for heuristic sweeps. The global procedure maintains a robust incumbent across all θ values and a global upper bound equal to the maximum of solved fixed- θ optima, pruned fixed- θ LP bounds, and unresolved fixed- θ branch-and-bound bounds. A θ value is pruned before branch-and-bound only if its valid fixed- θ LP upper bound is no larger than the global incumbent value. When a fixed- θ branch-and-bound run is stopped by a global incumbent cutoff (reported in implementations by statuses such as `cutoff_pruned`), the status is a global-pruning certificate for the robust search, not a claim that the fixed- θ subproblem has been solved locally to optimality.

The anytime gap semantics require one additional invariant. Before a finite global gap is reported, every $\theta \in \mathcal{B}$ must have a valid upper-bound record. The implementation initializes these records by computing the fixed- θ root LP upper bound for every candidate; rows with $C^\theta < 0$ or an infeasible root relaxation receive upper bound $-\infty$. If a row has no finite valid upper-bound record because of an implementation error, the global gap is reported as infinite or undefined rather than as a finite certificate.

Algorithm 2 Global θ -Enumerated Exact Robust Solver

1: **Input:** admissible menus, data (v_{ij}, s_{ij}, t_{ij}) , budget Γ , optional limits.
2: Construct $\mathcal{B} = \{0\} \cup \{|t_{ij}| : x_{i,j} \in X_i^{\text{ad}}\}$, deduplicate, and sort.
3: Initialize global incumbent with the best robust-feasible HullRound solution over $\mathcal{B}$.
4: For every $\theta \in \mathcal{B}$, initialize a valid upper-bound record: set it to $-\infty$ if $C^\theta < 0$, otherwise compute the fixed- θ hull-greedy root LP upper bound. Until all records are initialized, report no finite global gap.
5: **for** $\theta \in \mathcal{B}$ **do**
6: Build s_{ij}^θ , c_{ij}^θ , and C^θ .
7: **if** $C^\theta < 0$ **then**
8: mark θ infeasible and continue.
9: **end if**
10: Retrieve the initialized fixed- θ hull-greedy LP upper bound.
11: **if** the bound is below the global incumbent value **then**
12: prune θ and continue.
13: **end if**
14: Run fixed- θ hull-bound branch-and-bound for this θ , respecting remaining global time or node limits.
15: Update the global incumbent only if the returned selection passes the exact robust certificate.
16: Store the fixed- θ optimum or unresolved upper bound in the global upper-bound record.
17: **end for**
18: Certify the robust residual of the returned incumbent by (28).
19: **Output:** exact robust optimum if all relevant θ solves are certified; otherwise incumbent, global upper bound, and gap.

Theorem 14 (Global θ -enumerated exactness and anytime gap). *Let $\mathcal{B} = \{0\} \cup \{|t_{ij}| : i = 1, \dots, n, j \in J_i^{\text{ad}}\}$ be constructed from all original admissible options. If the fixed- θ branch-and-bound solver is run to optimality for every $\theta \in \mathcal{B}$ that is not validly pruned by a global LP-bound or incumbent cutoff, and θ values are pruned only when their valid fixed- θ LP upper bound is no larger than the global incumbent value, then Algorithm 2 returns an exact Γ -robust MCKP optimum. Under time or node limits, it returns a valid robust-feasible incumbent and a valid global upper bound, hence a valid global optimality gap, provided every $\theta \in \mathcal{B}$ has an initialized valid upper-bound record before a finite gap is reported. If any such record is missing or invalid, the finite global gap is not certified and is reported as infinite or undefined.*

Proof. Proposition 7 reduces the exact robust problem to the maximum over the finite family of fixed- θ MCKPs. Each unpruned fixed- θ optimum is certified by Theorem 13. A pruned θ cannot improve the global incumbent because its LP relaxation upper-bounds every integer solution at that θ , so such pruning is a valid global certificate even though it is not a local fixed- θ optimality statement. Taking the best certified incumbent over all unpruned candidates therefore gives the robust optimum. If limits stop some subproblems early, the incumbent remains feasible by the final robust certificate. Because every θ has an initialized valid upper-bound record, the maximum over solved optima, pruned LP bounds, active branch-and-bound bounds, and not-yet-searched root LP records upper-bounds the objective of every candidate in $\mathcal{B}$. This maximum therefore gives a valid global gap rather than an optimality claim. Without this per- θ upper-bound record invariant, a finite limited-run global gap would not be justified. $\square$

Lemma 15 (Safety of item-local hull reuse). *Fix an item and two θ candidates. If the transformed fixed- θ cost-value point multiset for that item is unchanged, then the item upper hull is identical to the hull obtained by independent recomputation and may be reused. If the point multiset changes, the hull must be rebuilt from the original admissible options. Since the exact fixed- θ branch-and-bound layer branches over the original admissible integer options and uses upper hulls only as LP-bound objects, this reuse rule cannot delete an integer candidate or make an LP upper bound invalid.*

Proposition 16 (Exactness of the full-breakpoint parametric θ sweep). *Let $\mathcal{B} = \{0\} \cup \{|t_{ij}| : i = 1, \dots, n, j \in J_i^{\text{ad}}\}$ be constructed from all original admissible options and sorted deterministically. In exact arithmetic, the full-breakpoint sweep visits exactly the same θ candidates as independent enumeration. At each $\theta \in \mathcal{B}$, the*

transformed fixed- θ data $(s^\theta, c^\theta, C^\theta)$ are the same as those obtained by independent recomputation. Item hulls are reused only when the induced cost-value point set and the integer-safe option set are unchanged; otherwise they are rebuilt. Therefore every fixed- θ LP bound, branch-and-bound result, incumbent, global upper bound, and global gap produced by the sweep has the same validity status as the corresponding independent full- θ enumeration. This is an exactness statement, not a worst-case runtime dominance claim.

Proof. The candidate construction is identical in the two procedures, so the same finite set $\mathcal{B}$ is visited. Between adjacent breakpoints, the affine identity $s_{ij}^\theta = s_{ij} - (|t_{ij}| - \theta)_+$ gives the incremental update used in the sweep. Recomputing the baselines, costs, and capacity from these updated values therefore yields exactly the same fixed- θ MCKP data as independent recomputation. If an item point set is unchanged, reusing its hull preserves the LP-bound object; if it changes, rebuilding gives the same hull that independent recomputation would construct by [Lemma 15](#). The branch-and-bound layer still branches over original integer-safe non-dominated options and uses hulls only for LP upper bounds. The global pruning and gap arguments are then those of [Theorem 14](#). $\square$

Implementation note. The proposition is an exact-arithmetic statement. In floating-point experiments, equality checks for sweep validation and hull reuse use stated tolerances, validation can be exhaustive or sampled, and every reported robust incumbent is checked against the original certificate ([28](#)).

6.6 Certificate semantics and implementation validation

The implementation records solver status, robust certificates, lower and upper bounds, gaps, node counts, and per- θ records. The same fixed- θ LP-bound routine is used for root θ pruning and node bounds, so limited runs report relaxation-based gaps, not heuristic estimates. HullRound supplies a robust-feasible incumbent and lower bound, but exactness comes only from closing every relevant fixed- θ search or validly pruning it. The validation suite compares against brute force on tiny instances; checks $\Gamma = 0$, $\Gamma = n$, all-zero uncertainty, infeasibility, binding residuals, duplicate θ values, node-limited anytime behavior, and below-hull optimal integer options; and compares certified benchmark rows with SCIP and HiGHS where available.

7 Numerical Results

We evaluate the structural and algorithmic claims in [Sections 4–6](#) on synthetic, pricing-derived, and semi-synthetic application instances. The numerical section is organized around certificate objects: (i) validity of the HullRound additive certificate and LP cross-checks, (ii) primal quality on rows with certified optima, (iii) exact certification status and tight-capacity failure modes, (iv) full-breakpoint sweep equivalence and overhead, and (v) the semi-synthetic pricing frontier as a decision demonstration. Extended stress grids, open-solver scaling diagnostics, and claim-to-evidence tables are collected in the appendices. Unless otherwise stated, curves show medians across independent seeds and shaded bands or error bars show interquartile ranges.

7.1 Experimental protocol

The experiments use synthetic economic-pricing instances, targeted synthetic stress families, and the semi-synthetic pricing generator in [Section 8](#). In the economic generator, reference prices, exposures, baseline demands, and elasticities are sampled independently; menus are reference-band finite menus, nominal demand is own-price linear in the relative price movement, uncertainty is proportional to nominal demand, and the margin target is recalibrated with a small baseline slack. The targeted families vary hull compression, capacity tightness, boundary budgets, and the number of distinct θ values.

Nested prefixes isolate size effects and independent seeds quantify sampling variation. The exact benchmarks are deliberately smaller than the HullRound scaling runs because each candidate θ must be certified or validly bounded. For every retained solution we reconstruct the selected menu options and recompute the exact robust residual

$$Z_{\Gamma}(\mathbf{x}) = \sum_i s_i(x_i) - \sum_{k=1}^{\Gamma} |t|_{(k)}(\mathbf{x}).$$

The validation script also checks that the hull-greedy LP has at most one fractional item, verifies $L_{\text{rd}} \leq \Delta V_{\text{max}}^{\theta} + 10^{-7}$ on all reported gap runs, and cross-checks the hull-greedy LP against HiGHS on a sample of fixed- θ LP relaxations. The completed run has 0 bound violations; the maximum LP cross-check residual is below 10^{-9} .

The reported benchmark records were generated on macOS arm64 (Apple M4, 10 CPU cores). The recorded software stack was Python 3.14.2 with NumPy 2.4.4, SciPy 1.17.1, Matplotlib 3.10.9, PySCIPOpt 6.1.0 with SCIP 10.0.1, and highspy/HiGHS 1.14.0. SCIP calls set the available thread parameter to one, and the specialized Python solvers and HiGHS runs are executed as single-process runs. Commercial solvers such as Gurobi and CPLEX were unavailable and are not benchmarked; SCIP was available and HiGHS was available. A row is *certified* only when the exact solver or MILP baseline reports optimality. A HullRound row marked *feasible* means exact residual feasibility, not optimality. A *limited* exact row reports a valid incumbent–upper-bound gap under a time or node limit. Entries marked n/r were not run and are not failures. The relative gap is

$$(\text{UB} - \text{LB}) / \max\{1, |\text{UB}|\},$$

unless the generating script records the equivalent solver convention. The direct robust MILP uses the Bertsimas–Sim linearization; θ enumeration certifies the global robust discrete optimum only when every fixed- θ subproblem is certified optimal.

7.2 LP relaxation and integrality-gap validation

Figure 1 validates the additive rounding theorem on independent replicated instances. Panel (a) reports the round-down loss divided by $\Delta V_{\text{max}}^{\theta}$ before optional repair/completion. Across all replicated gap rows, the median ratio is 0.033, the 95th percentile is 0.188, and the maximum is 0.547; no run violates the bound. Panels (b)–(c) add a larger- n check at $\Gamma = \lfloor \sqrt{n} \rfloor$ through $n = 500$. The median relative gap at the largest tested size is 0.005%, and $n \times \text{Gap}_{\text{LP}}$ remains bounded over the tested sizes, consistent with the one-item additive scale in Proposition 11. We do not infer an asymptotic rate from these finite samples.

7.3 Certified benchmark for the exact solver

Table 2 summarizes the certified exact-solver benchmark. The benchmark contains 64 instances and 320 method rows across six synthetic families plus a capped economic extension. We use it to check certificate semantics and failure modes, not to claim general-purpose runtime dominance. The comparison includes HullRound, two exact θ -B&B configurations, and direct robust MILP baselines in SCIP and HiGHS. All exact θ -B&B configurations enumerate the full candidate set $\mathcal{B} = \{0\} \cup \{|t_{ij}|\}$ and preserve the original non-dominated integer option sets; upper hulls are used only for LP upper bounds. Appendix C reports the smaller open-solver cross-checks and scalability diagnostics that are not needed for the main certification claim.

The optimized exact configuration uses exact-safe caches, incumbent cutoffs, and LP-bound-descending θ ordering. These changes reduce median runtime from 0.3476s to 0.2830s and median node count from 87 to 49 relative to the uncached increasing- θ baseline, but SCIP and HiGHS match the optimized exact solver on all 56 rows where all methods certify optimality and are faster on the median row in this execution

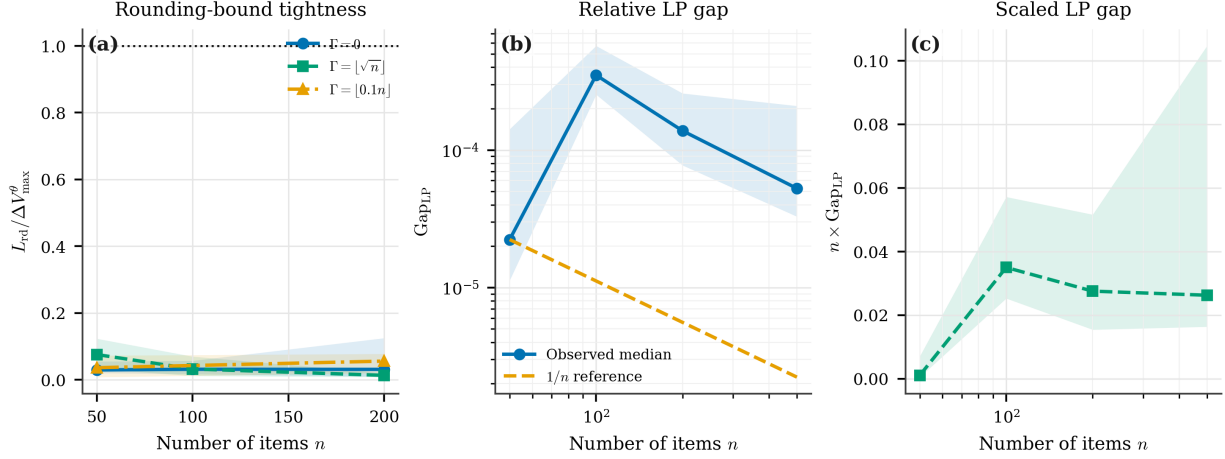


Figure 1. Replicated integrality-gap validation. Panel (a) shows $L_{rd} / \Delta V_{\max}^{\theta}$ with the theoretical ceiling at one on 48 replicated rows; the ceiling is never crossed and the maximum observed ratio is 0.547. Panel (b) reports the larger- n relative LP gap with a $1/n$ reference line, and panel (c) shows $n \times \text{Gap}_{LP}$ on 12 scaling rows. Lines are medians and bands are interquartile ranges over independent seeds.

Table 2. Certified benchmark snapshot. “Cert./feas.” means exact optimality for exact methods and robust-feasible incumbency for HullRound. HullRound rows are not exact optimality claims.

Method	Rows	Cert./feas.	Limited	Median time (s)	Median nodes	Median gap
HullRound	64	64	0	0.0269	—	—
Global θ -B&B	64	56	8	0.3476	87	0.0000
Global θ -B&B + cache/cutoff/order	64	56	8	0.2830	49	0.0000
SCIP robust MILP	64	56	8	0.0040	—	0.0000
HiGHS robust MILP	64	56	8	0.0066	—	0.0000

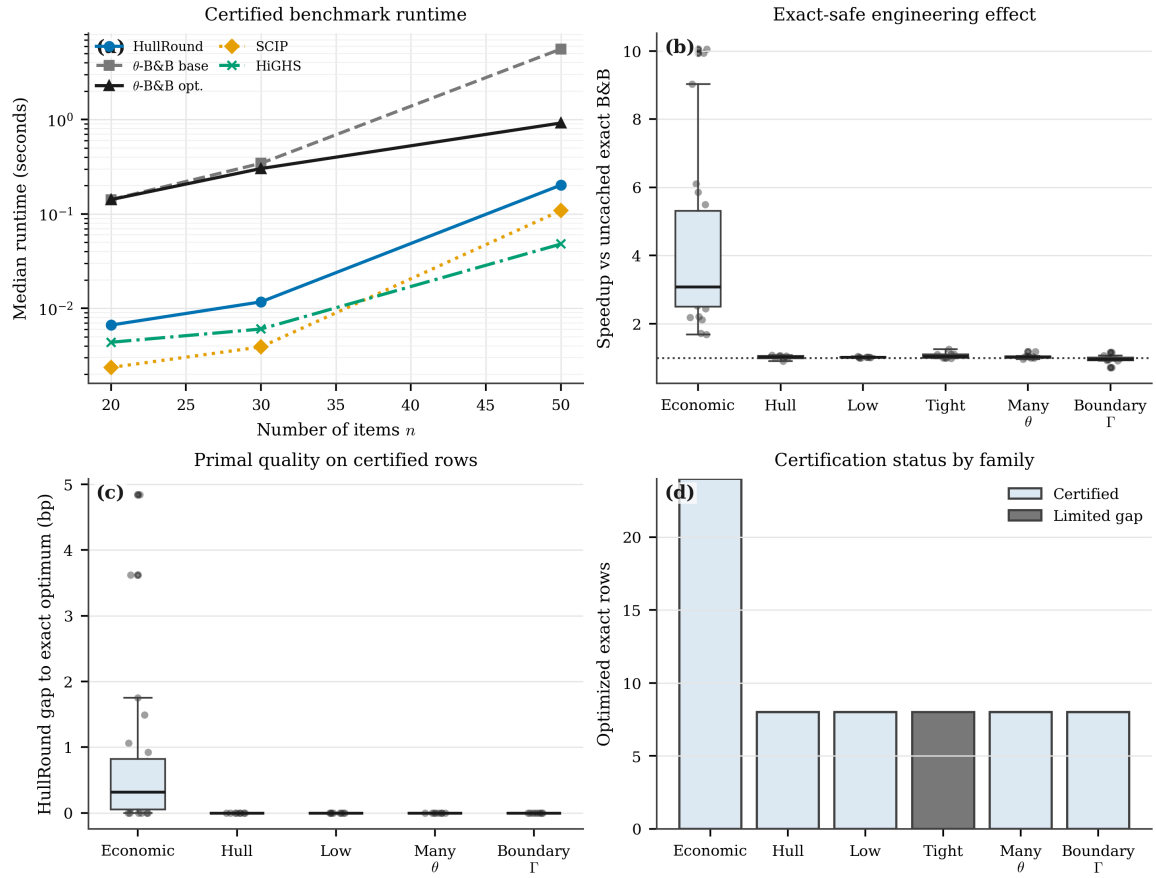


Figure 2. Certified benchmark diagnostics. The certification panels are the primary evidence: panel (d) separates certified exact rows from limited valid-gap rows by family, and panel (c) reports HullRound objective gaps to certified exact optima in basis points. Panel (b) shows the exact-safe engineering effect relative to the uncached exact baseline. Panel (a) reports runtime only as a baseline sanity check; generic MILP solvers are strong runtime baselines, while the specialized method is evaluated for certificate semantics and failure diagnosis.

environment. The value of the specialized run is therefore the explicit record of which θ subproblems are infeasible, pruned, solved, or left with valid upper-bound gaps.

Table C.4 breaks the optimized exact solver down by instance family. The eight unresolved exact rows all come from the tight-capacity family; limited runs return robust-feasible incumbents and valid upper-bound gaps instead of optimality claims. The non-tight families are certified in this benchmark, often after pruning nearly all θ candidates by valid fixed- θ LP upper bounds. The tight-capacity family is therefore the principal stress test for future branch selection, cutoffs, and incremental bound maintenance.

HullRound returns robust-feasible solutions on all 64 rows. On the 56 rows where the optimized exact solver certifies optimality, HullRound’s median relative gap to the exact optimum is zero, and the maximum relative gap is 4.9×10^{-4} . Figure 2(c) shows the gap distribution in basis points. This supports HullRound’s role as a strong primal heuristic and practical baseline even though it is not an exact solver.

Table C.5 reports a targeted follow-up on many- θ and tight-capacity regimes. The many- θ family completed on all 24 instances with matching optima across optimized θ -B&B, SCIP, and HiGHS. The tight-capacity run was stopped after three instances because exact B&B and the open-source MILP baselines were already consuming the per-method limit; the limited exact B&B rows still returned relative gaps below 1.5×10^{-3} .

7.4 Parametric sweep: exactness, maintenance, and regime-dependent payoff

Table 3–Table C.7 report a bounded final sweep campaign. The ablation uses eight construction cases per family, and the solver comparison uses four matched cases per family with a 20-second limit and a 120,000-node limit. All exact rows in the comparison are certified. The sweep uses the same full breakpoint set as independent enumeration, validates sampled fixed- θ states against independent recomputation, reuses item hulls only under the point-set criterion in Lemma 15, and validates every accepted incumbent by the original robust certificate.

Table 3. Exact sweep validation. The table reports four matched solver cases per family. Validation is sampled fixed- θ recomputation; “Obj. matches” compares exact enumeration and the sweep on certified matched rows.

Family	Inst.	Med. $ \mathcal{B} $	IQR $ \mathcal{B} $	Enum cert.	Sweep cert.	Obj. matches	Validation	Cert. fail	Med. gap
many- θ	4	351	351–351	4/4	4/4	4/4	4/4	0	0.0000
low compression	4	351	351–351	4/4	4/4	4/4	4/4	0	0.0000
tight capacity	4	8.0	8.0–8.0	4/4	4/4	4/4	4/4	0	0.0000
non-tight control	4	4.0	4.0–4.0	4/4	4/4	4/4	4/4	0	0.0000

The results support the exactness and traceability claims but do not support a broad runtime-payoff claim. The sweep matches independent full- θ enumeration on all 16 certified matched rows and has no robust-certificate failures in this campaign. Safe item-local reuse is meaningful in the many- θ family, where the median reuse rate is 0.489, but geometry changes frequently in the low-compression, tight-capacity, and non-tight-control families. As a result, validation and maintenance overhead offset reuse in the bounded run: Figure 3 shows median sweep/enumeration runtime ratios above one in every tested family. Thus the present sweep is best interpreted as an exact certifying and diagnostic architecture; stronger interval bounds, cuts, and lower-overhead validation would be needed before making a performance-algorithm claim.

This benchmark sharpens the interpretation of the exact solver. The specialized implementation provides an independent certifying backbone, full-breakpoint traceability, safe hull-maintenance diagnostics, and valid gap accounting. Tight-capacity rows remain hard because residual LP-bound computation is the main node-time cost, cutoff pruning is weak when incumbents are close but not strong enough to close the bound, branch selection still leaves large residual trees, and exact global mode must account for the full θ candidate set. These rows motivate stronger pruning, cuts, incumbent generation, and larger matched solver campaigns.

Table C.8 reports a non-tight stress test focused on HullRound quality. The experiment uses the economic, many- θ , hull-compression, and low-compression families with $n \in \{50, 80\}$, $m \in \{8, 10\}$, $\Gamma \in \{\lfloor \sqrt{n} \rfloor, \lfloor 0.1n \rfloor\}$, two seeds, and a 45-second limit for exact methods. All 64 optimized θ -B&B rows are certified, and SCIP and HiGHS match the exact objective on all 64 certified rows. HullRound is robust-feasible on all rows; its median relative gap to the certified exact optimum is zero, and its maximum relative gap is 4.74×10^{-5} .

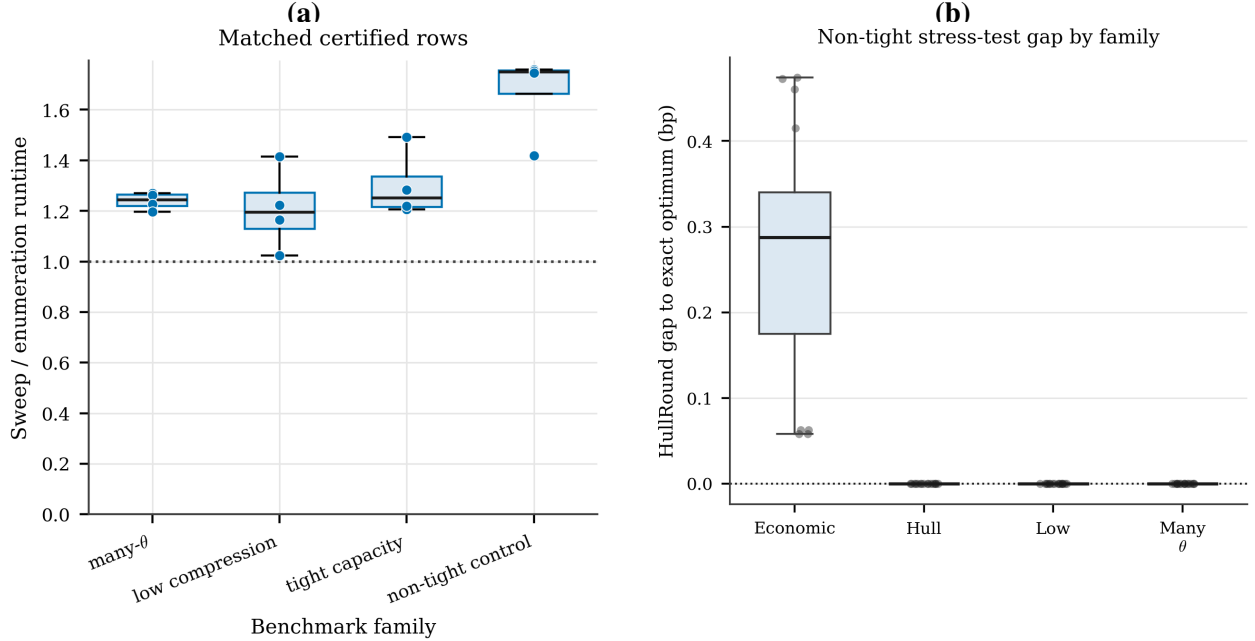


Figure 3. Diagnostic limitations and primal quality. Panel (a) reports matched sweep payoff by regime: ratios are exact-sweep runtime divided by independent full- θ enumeration runtime on certified matched rows, with parity marked by the dotted line. Panel (b) reports HullRound gap to certified exact optimum in the non-tight stress test; nonzero gaps occur only in the economic family and remain below one basis point.

The full claim-to-evidence table is deferred to Table C.9. The main text reports only the headline quantities needed to interpret the algorithms.

7.5 Experimental limitations

The experiments validate algorithmic structure, not an empirical demand specification. They use synthetic and semi-synthetic own-price models, no cross-price substitution, and constructed uncertainty bands. The exact solver is a correctness-first Python implementation and remains limited by full θ enumeration, residual LP-bound evaluation, incumbent/cutoff strength, and branch selection, especially on tight-capacity rows. No Gurobi or CPLEX comparison is included because licenses were unavailable; in the certified benchmark, direct SCIP/HiGHS dominate the median runtime.

8 Semi-Synthetic Application: Robust Category Pricing

8.1 Calibration and portfolio generator

The application is a controlled decision-frontier demonstration for the robust-MCKP framework. It asks how the robustness budget changes revenue, deterministic protection, stress-test behavior, and price movement

under a reproducible finite-menu generator. It uses documented semi-synthetic calibration defaults rather than observed transaction data. The generator creates eight $n = 180$, $m = 12$ portfolios with five product segments, segment-specific reference-price bands, psychologically rounded menu prices, own-price iso-elastic demand, proportional uncertainty bands, and a margin target calibrated to a small nominal reference slack. The design is controlled and reproducible, not causal demand estimation.

8.2 Experimental design

The application compares the nominal $\Gamma = 0$ HullRound policy, the robust HullRound frontier, and the box-robust $\Gamma = n$ policy. We evaluate

$$\Gamma \in \{0, 5, \lfloor \sqrt{180} \rfloor, \lfloor 0.1n \rfloor, \lfloor 0.25n \rfloor, n\} = \{0, 5, 13, 18, 45, 180\}.$$

For each policy we record normalized revenue, the exact robust residual $Z_\Gamma(\mathbf{x})$, violation probabilities under 2500 stress scenarios, mean margin shortfall, price changes relative to the same seed’s nominal policy, segment-level price movements, and positive margin-concentration diagnostics. Stress protocols include iid, common-factor, segment-block, heavy-tailed, and under-calibrated uncertainty shocks. A separate exact small-subset check uses $n = 50$, $m = 10$, two seeds, and $\Gamma \in \{\lfloor \sqrt{n} \rfloor, \lfloor 0.1n \rfloor\}$ to compare HullRound with exact global θ -B&B. The study is therefore a controlled decision experiment rather than transaction-level scanner-data validation, causal demand estimation, or evidence about cross-price substitution.

8.3 Robustness frontier

Table 4 and Figure 4 show the resulting revenue–robustness frontier. Moving from $\Gamma = 0$ to $\Gamma = 13 = \lfloor \sqrt{180} \rfloor$ reduces the median revenue ratio from 1.0000 to 0.9968, changes a median 11.7% of prices, and reduces the median iid violation rate from 0.499 to zero. Segment-block shocks are harder: their median violation rate falls from 0.493 at $\Gamma = 0$ to 0.098 at $\Gamma = 13$, and to 0.004 at $\Gamma = 45$. Full box protection drives the reported stress violations to zero in this run, but at a larger median revenue loss of about 0.85% and with a median 25.0% of prices changed. These results support the use of Γ as a tunable revenue–robustness control rather than a recommendation for box protection.

Table 4. Semi-synthetic category-pricing frontier for HullRound. Revenue is normalized by the same seed’s nominal $\Gamma = 0$ solution. Violation probabilities are diagnostic stress-test frequencies outside the robust optimization model. The certificate column is the deterministic robust residual in native residual units, rounded as generated by the summary script.

Γ	Rev.	Cert.	IID viol.	Block viol.	Heavy viol.	Changed	$ \Delta p $
0	1.0000	0.0	0.499	0.493	0.497	0.000	0.000
5	0.9981	0.0	0.002	0.222	0.033	0.075	0.020
13	0.9968	0.0	0.000	0.098	0.001	0.117	0.028
18	0.9962	0.0	0.000	0.059	0.000	0.142	0.032
45	0.9939	0.0	0.000	0.004	0.000	0.183	0.054
180	0.9915	0.0	0.000	0.000	0.000	0.250	0.094

8.4 Application takeaways

Figure 4(e)–(f) and Table D.1 show that the controlled frontier is not implemented by a uniform markup or markdown: at $\Gamma = 13$, a minority of SKUs move within their admissible bands, with segment effects that reflect both uncertainty and volume. The exact small-subset check in Table D.2 is a quality check on the

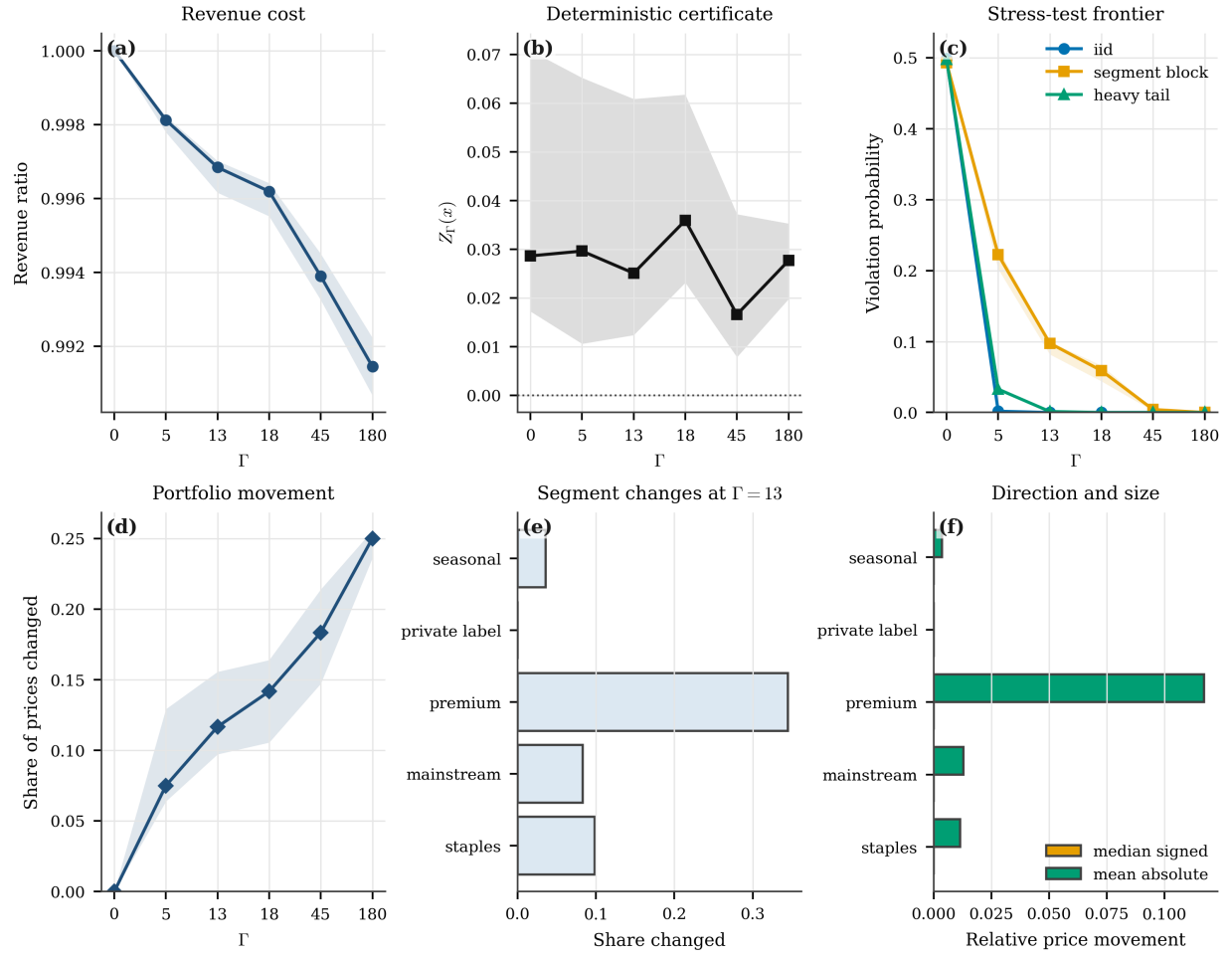


Figure 4. Semi-synthetic category-pricing diagnostics. Panels (a)–(d) report median normalized revenue, deterministic robust residual, stress-test violation rates, and share of changed prices across eight independent portfolios. Panels (e)–(f) summarize segment-level price-movement diagnostics at $\Gamma = 13$. Shaded bands in line plots are interquartile ranges over seeds.

generator, not a large benchmark; on its four certified rows, HullRound is robust-feasible with median relative gap 1.16×10^{-5} and maximum gap 1.71×10^{-5} to exact global θ -B&B. The application should be read as a reproducible frontier tool for choosing Γ by inspecting certificates, stress-test violations, revenue loss, and price movement. Deployment would require demand estimation, uncertainty calibration, and out-of-sample validation on observed data.

A visual synthesis plot is reported in [Appendix C.2](#). It is not a separate claim; it summarizes the same diagnostic benchmark records used in the main numerical tables.

9 Conclusion

This paper develops a certifying decomposition for finite-menu integer-budget Γ -robust MCKP. The robust capacity constraint is reduced exactly to the full original breakpoint set in θ ; each fixed- θ subproblem is an ordinary MCKP with an explicit upper-hull greedy LP bound and an optimal LP solution with at most one fractionally mixed class. This yields the robust-feasible HullRound certificate and an exact θ -enumerated branch-and-bound layer that preserves below-hull non-dominated integer options and reports valid gaps under limits.

The computational evidence supports this role. HullRound is a low-overhead robust-feasible primal method with very small gaps on certified non-tight instances, while global θ -B&B provides independent exact certificates or valid limited-run gaps. The parametric sweep validates full-breakpoint equivalence and safe item-local maintenance, but in the bounded tested regimes it is diagnostic rather than a broad runtime improvement. Generic MILP solvers remain strong runtime baselines.

The remaining limitations are modeling and exact-search scope: the experiments are synthetic or semi-synthetic, the pricing illustration is own-price and separable, and tight-capacity rows remain difficult for the exact prototype. Future work should study stronger pruning and incumbent generation, richer demand and uncertainty models, commercial-solver comparisons when licenses are available, and validation on observed pricing data.

Data and code availability

To facilitate reproducibility, the solver package, tests, experiment drivers, plotting and summarization scripts, and reproducibility instructions are maintained on GitHub:

https://github.com/eric939/robust_mckp

The public repository was reachable during this build; the repository HEAD used for the code-availability check was 2e2829e55243b846e5944965a839eb9b28dd5e9c. The exact manuscript-generation snapshot for this submission, including generated tables, figures, auto-number files, and result summaries, is supplied with the submission package. The semi-synthetic application stores only generated calibration and result records; no raw public transaction data are redistributed.

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References

- Ben-Tal, A. and Nemirovski, A. (1999). Robust solutions of uncertain linear programs. *Operations Research Letters*, 25(1), 1–13. [https://doi.org/10.1016/S0167-6377\(99\)00016-4](https://doi.org/10.1016/S0167-6377(99)00016-4).
- Bertsimas, D. and Sim, M. (2004). The price of robustness. *Operations Research*, 52(1), 35–53. <https://doi.org/10.1287/opre.1030.0065>.
- Boukhari, S. and Hifi, M. (2023). An exact algorithm for the multiple-choice knapsack problem with setups. *Computers & Industrial Engineering*, 188, 109818. <https://doi.org/10.1016/j.cie.2023.109818>.
- Boukhari, S. and Hifi, M. (2024). Combining local branching and descent method for solving the multiple-choice knapsack problem with setups. *International Transactions in Operational Research*, 31(1), 29–52. <https://doi.org/10.1111/itor.13326>.
- Caserta, M. and Voß, S. (2019). The robust multiple-choice multidimensional knapsack problem. *Omega*, 86, 16–27. <https://doi.org/10.1016/j.omega.2018.06.014>.
- Dyer, M. E. (1984). An $O(n)$ algorithm for the multiple-choice knapsack linear program. *Mathematical Programming*, 29(1), 57–63. <https://doi.org/10.1007/BF02591729>.
- Goerigk, M., Lendl, S., and Wulf, L. (2021). Robust combinatorial optimization with locally budgeted uncertainty. *Open Journal of Mathematical Optimization*, 2, Article 3. <https://doi.org/10.5802/ojmo.5>.
- Hamzeei, M., Lim, A., and Xu, J. (2022). Robust price optimization of multiple products under interval uncertainties. *Journal of Revenue and Pricing Management*, 21(4), 442–454. <https://doi.org/10.1057/s41272-021-00351-w>.
- Ibaraki, T., Hasegawa, T., Teranaka, K., and Iwase, J. (1978). The multiple-choice knapsack problem. *Journal of the Operations Research Society of Japan*, 21(1), 59–95. <https://doi.org/10.15807/jorsj.21.59>.
- Kellerer, H., Pferschy, U., and Pisinger, D. (2004). *Knapsack Problems*. Springer.
- Monaci, M., Pferschy, U., and Serafini, P. (2013). Exact solution of the robust knapsack problem. *Computers & Operations Research*, 40(11), 2625–2631. <https://doi.org/10.1016/j.cor.2013.05.005>.
- Nauss, R. M. (1978). The 0–1 knapsack problem with multiple choice constraints. *European Journal of Operational Research*, 2(2), 125–131. [https://doi.org/10.1016/0377-2217\(78\)90108-X](https://doi.org/10.1016/0377-2217(78)90108-X).
- Pisinger, D. (1995). A minimal algorithm for the multiple-choice knapsack problem. *European Journal of Operational Research*, 83(2), 394–410. [https://doi.org/10.1016/0377-2217\(95\)00015-I](https://doi.org/10.1016/0377-2217(95)00015-I).
- Shao, Z. Y. E. (2026). Robust discrete pricing optimization via multiple-choice knapsack reductions. arXiv:2603.18653 [math.OC]. <https://arxiv.org/abs/2603.18653>.
- Sinha, P. and Zoltners, A. A. (1979). The multiple-choice knapsack problem. *Operations Research*, 27(3), 503–515. <https://doi.org/10.1287/opre.27.3.503>.

A Proof Details

We provide the first three steps of the proof of [Theorem 5](#) for completeness. These arguments are standard; we include them because our notation differs from prior sources.

Step 1: Convex-hull reformulation. Fix an item i . For any feasible vector $(z_{ij})_{j=1}^m$ with $\sum_j z_{ij} = 1$ and $z_{ij} \geq 0$, define

$$(c_i, v_i) := \left(\sum_{j=1}^m c_{ij} z_{ij}, \sum_{j=1}^m v_{ij} z_{ij} \right).$$

Then $(c_i, v_i) \in \text{conv}(\mathcal{P}_i)$ by definition of convex hull, since it is a convex combination of points in $\mathcal{P}_i$. Conversely, every point in $\text{conv}(\mathcal{P}_i)$ admits such a representation as a convex combination of points in $\mathcal{P}_i$. Therefore, (17) is equivalent to choosing $(c_i, v_i) \in \text{conv}(\mathcal{P}_i)$ for all i to maximize $\sum_i v_i$ subject to $\sum_i c_i \leq C$.

Step 2: Restriction to the upper hull (proof of (i)). Fix i and a feasible solution with induced $(c_i, v_i) \in \text{conv}(\mathcal{P}_i)$. Consider the set

$$U_i(c) := \sup\{v : (c, v) \in \text{conv}(\mathcal{P}_i)\}.$$

Since $\text{conv}(\mathcal{P}_i)$ is a compact polytope, the supremum is attained and $U_i(c)$ describes the upper hull of $\text{conv}(\mathcal{P}_i)$, equivalently the upper envelope of the convex hull. Let $v'_i := U_i(c_i)$. Then $(c_i, v'_i) \in \text{conv}(\mathcal{P}_i)$, and $v'_i \geq v_i$ by construction. Replacing (c_i, v_i) by (c_i, v'_i) for each i preserves feasibility of $\sum_i c_i \leq C$ and weakly increases the objective. Hence replacing a point by the envelope is LP-safe, and there exists an optimal solution with each (c_i, v_i) on the upper hull.

The upper hull of a polytope in $\mathbb{R}^2$ is a polyline connecting consecutive vertices of $\mathcal{H}_i$. Thus, if (c_i, v_i) lies on the hull and $c_i \in [c_{i,k}^H, c_{i,k+1}^H]$ for some k , then (c_i, v_i) is a convex combination of the two adjacent vertices $(c_{i,k}^H, v_{i,k}^H)$ and $(c_{i,k+1}^H, v_{i,k+1}^H)$. This proves (i).

Step 3: Segment-length parameterization (proof of (ii)). Fix an item i and let the efficient upper-hull vertices after equal-cost merging and dominance cleanup be

$$\mathcal{H}_i = \{(c_{i,1}^H, v_{i,1}^H), \dots, (c_{i,p_i}^H, v_{i,p_i}^H)\}, \quad c_{i,1}^H < \dots < c_{i,p_i}^H, \quad v_{i,1}^H < \dots < v_{i,p_i}^H.$$

For $k = 1, \dots, p_i - 1$ define

$$\Delta c_{i,k}^H := c_{i,k+1}^H - c_{i,k}^H > 0, \quad \rho_{i,k}^H := \frac{v_{i,k+1}^H - v_{i,k}^H}{\Delta c_{i,k}^H}.$$

If $p_i = 1$, the only upper-hull point is $(c_{i,1}^H, v_{i,1}^H)$ and there are no segment variables. Otherwise, every point (c_i, v_i) on the upper hull lies on a segment between adjacent vertices, so there exist $k^* \in \{1, \dots, p_i - 1\}$ and $\alpha \in [0, 1]$ such that

$$(c_i, v_i) = (1 - \alpha)(c_{i,k^*}^H, v_{i,k^*}^H) + \alpha(c_{i,k^*+1}^H, v_{i,k^*+1}^H).$$

For $p_i > 1$, define segment-length variables $\ell_{i,k}$ by

$$\ell_{i,k} := \begin{cases} \Delta c_{i,k}^H, & k < k^*, \\ \alpha \Delta c_{i,k^*}^H, & k = k^*, \\ 0, & k > k^*, \end{cases} \quad \text{so that} \quad 0 \leq \ell_{i,k} \leq \Delta c_{i,k}^H,$$

and, for $k = 1, \dots, p_i - 2$,

$$\ell_{i,k+1} > 0 \Rightarrow \ell_{i,k} = \Delta c_{i,k}^H.$$

Then the identities

$$c_i = c_{i,1}^H + \sum_{k=1}^{p_i-1} \ell_{i,k}, \quad v_i = v_{i,1}^H + \sum_{k=1}^{p_i-1} \rho_{i,k}^H \ell_{i,k}$$

hold, since on each segment k the increase in v equals $\rho_{i,k}^H$ times the increase in c . Summing these expressions over i and substituting into the convex-hull formulation from Step 1 yields

$$\begin{aligned}
\max \quad & \sum_{i=1}^n \left(v_{i,1}^H + \sum_{k=1}^{p_i-1} \rho_{i,k}^H \ell_{i,k} \right) \\
\text{s.t.} \quad & \sum_{i=1}^n \left(c_{i,1}^H + \sum_{k=1}^{p_i-1} \ell_{i,k} \right) \leq C, \\
& 0 \leq \ell_{i,k} \leq \Delta c_{i,k}^H, \\
& \ell_{i,k+1} > 0 \Rightarrow \ell_{i,k} = \Delta c_{i,k}^H \quad (k = 1, \dots, p_i - 2).
\end{aligned}$$

Under $C \geq \sum_i c_{i,1}^H$, moving $\sum_i c_{i,1}^H$ to the right-hand side gives (18); if $C < \sum_i c_{i,1}^H$, the LP is infeasible.

It remains to justify dropping the prefix constraints when computing the segment-knapsack value. The efficient upper envelope is concave, so within each item the slopes $\rho_{i,k}^H$ are nonincreasing in k . Consider an optimum of the relaxed segment problem that uses a later segment $k+1$ of item i while an earlier segment k is not saturated. Moving an amount of capacity $\varepsilon \leq \min\{\ell_{i,k+1}, \Delta c_{i,k}^H - \ell_{i,k}\}$ from segment $k+1$ to segment k weakly improves the objective because $\rho_{i,k}^H \geq \rho_{i,k+1}^H$. Repeating this exchange restores prefix order without lowering value. For equal slopes, choosing earlier segments before later segments gives a prefix-feasible optimum directly. Thus the relaxed continuous segment knapsack has an optimal prefix-feasible solution, proving (ii).

B Implementation Validation

The implementation validation is designed to check certificate semantics rather than only numerical performance. The unit tests compare tiny instances against brute force, verify $\Gamma = 0$ and $\Gamma = n$, all-zero uncertainty, infeasible cases, binding residuals, duplicate θ values, below-hull integer optima, and node-limited valid-gap behavior. The final robust residual is recomputed from the original selection for every retained incumbent, so Monte Carlo stress tests are diagnostics only and never replace the deterministic certificate $Z_\Gamma(\mathbf{x}) \geq 0$. The publication scripts also cross-check sampled fixed- θ LP relaxations against HiGHS and record whether SCIP/HiGHS baselines are available in the execution environment.

Table B.1. Executable certification-risk checks included in the regression suite. These tiny cases target the three structural pitfalls most likely to invalidate the certificate interpretation.

Risk checked	Constructed validation	Regression test
Reduced θ set can falsely reject a feasible robust selection	Two selected uncertainty magnitudes equal one with $s_1 = s_2 = 0.75$ and $\Gamma = 1$: checking only $\theta = 0$ rejects, while the full set includes $\theta = 1$ and certifies feasibility.	<code>tests/test_global_theta_bnb.py</code> : reduced- θ unsafe case
Upper-hull filtering is not integer-safe	One group has points $(0, 0)$, $(5, 8)$, $(10, 20)$ at capacity five; the middle point is below the upper hull but is the integer optimum.	<code>tests/test_exact_bnb.py</code> and <code>tests/test_global_theta_bnb.py</code> : below-hull optimum cases
Finite limited-run gaps require per- θ upper-bound records	A node-limited global run initializes a valid root upper-bound record for every full-breakpoint candidate before reporting any finite gap.	<code>tests/test_global_theta_bnb.py</code> : per- θ upper-bound-record case

C Extended Numerical Tables

This appendix collects the secondary numerical figures and tables used for auditability. They preserve generated artifacts and labels for reproducibility, but the main claims are the four empirical objects reported in [Sections 7](#) and [8](#).

Table C.1. Extended rounding diagnostics. Objective losses are measured relative to the completed HullRound solution on the same instance.

Variant	Median loss vs. HR	Median certificate
Round-down	0.004%	22.03
Repair	0.002%	13.10
No completion	0.001%	6.27
HullRound	0.000%	0.11

Table C.2. Small exact benchmark with open solvers. “Full” is the direct Bertsimas–Sim robust MILP; “Enum” is exact enumeration over feasible θ values with a fixed- θ MCKP MILP. Certified optima are used only when every required solve reports optimality.

n	Γ	cert.	med. gap	max gap	HullRound s	SCIP full s	SCIP enum s	HiGHS enum s
30	$\lfloor 0.1n \rfloor$	5/5	0.00%	0.01%	0.11	0.05	10.14	8.16
30	$\lfloor \sqrt{n} \rfloor$	5/5	0.00%	0.01%	0.11	0.06	9.93	7.43
30	0	5/5	0.00%	0.01%	0.10	0.03	8.73	7.13

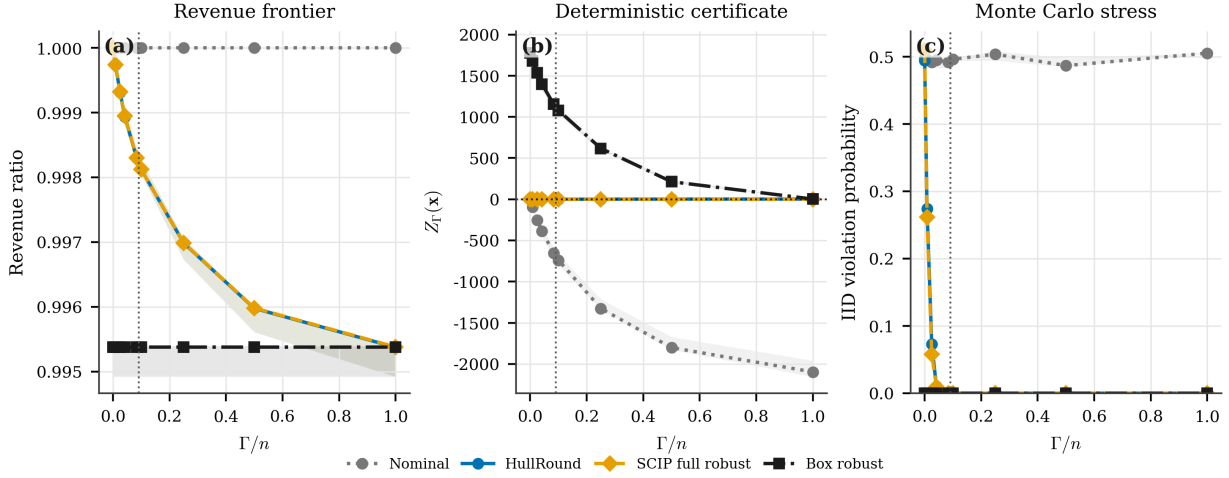


Figure C.1. Method-level robustness frontier on 27 synthetic method-frontier instances. Panel (a) reports normalized revenue, panel (b) the exact worst-case residual $Z_{\Gamma}(\mathbf{x})$, and panel (c) iid Monte Carlo violation probability. Lines are medians over three seeds, bands are interquartile ranges, and the vertical dotted line marks $\Gamma = \lfloor \sqrt{n} \rfloor$. The deterministic certificate, not the simulation, is the robustness proof.

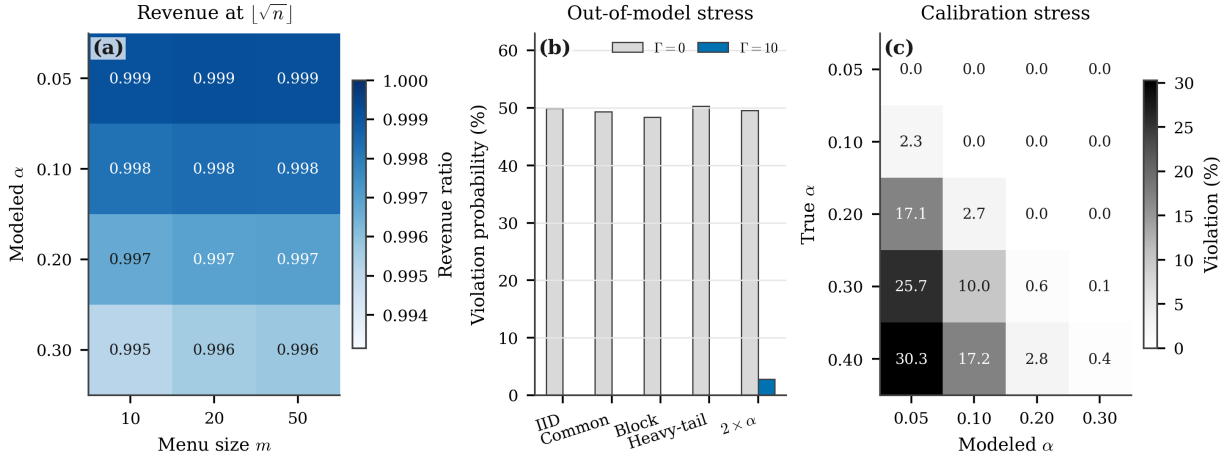


Figure C.2. Extended robustness-design diagnostics. Panel (a) reports median revenue ratios on a 120-row sensitivity grid relative to each seed’s $\Gamma = 0$ solution; all cells are exactly robust-certified for their stated Γ . Panel (b) reports Monte Carlo violation probabilities under stochastic protocols not used by the optimization model. Panel (c) reports iid violation rates when the true uncertainty scale differs from the modeled scale.

Table C.3. Scalability summary. “SCIP opt.” counts certified direct full-robust MILP solves; “HiGHS fixed” counts certified fixed- θ solves at the HullRound-selected θ . “n/r” means not run and is not counted as a failure.

n	m	HullRound s	median Z	SCIP opt.	SCIP gap	SCIP s	HiGHS fixed
50	10	0.32	0.8	3/3	0.000	0.25	3/3
100	10	1.18	1.3	3/3	0.000	0.48	3/3
200	10	4.97	0.8	3/3	0.000	1.50	3/3
500	10	31.57	0.2	n/r	—	—	n/r

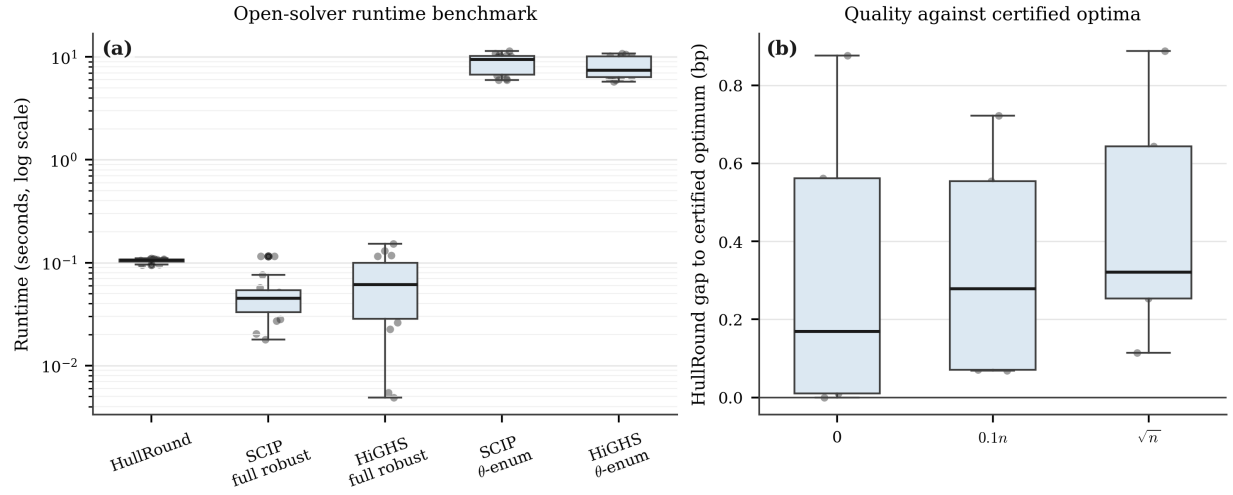


Figure C.3. Open-solver benchmark in the execution environment. Panel (a) compares runtime distributions on the small certified benchmark; panel (b) reports HullRound gaps to certified optima by Γ regime on 15 matched certified instances. SCIP/PySCIPOpt and HiGHS are used on rows where they are run; commercial solvers are not benchmarked.

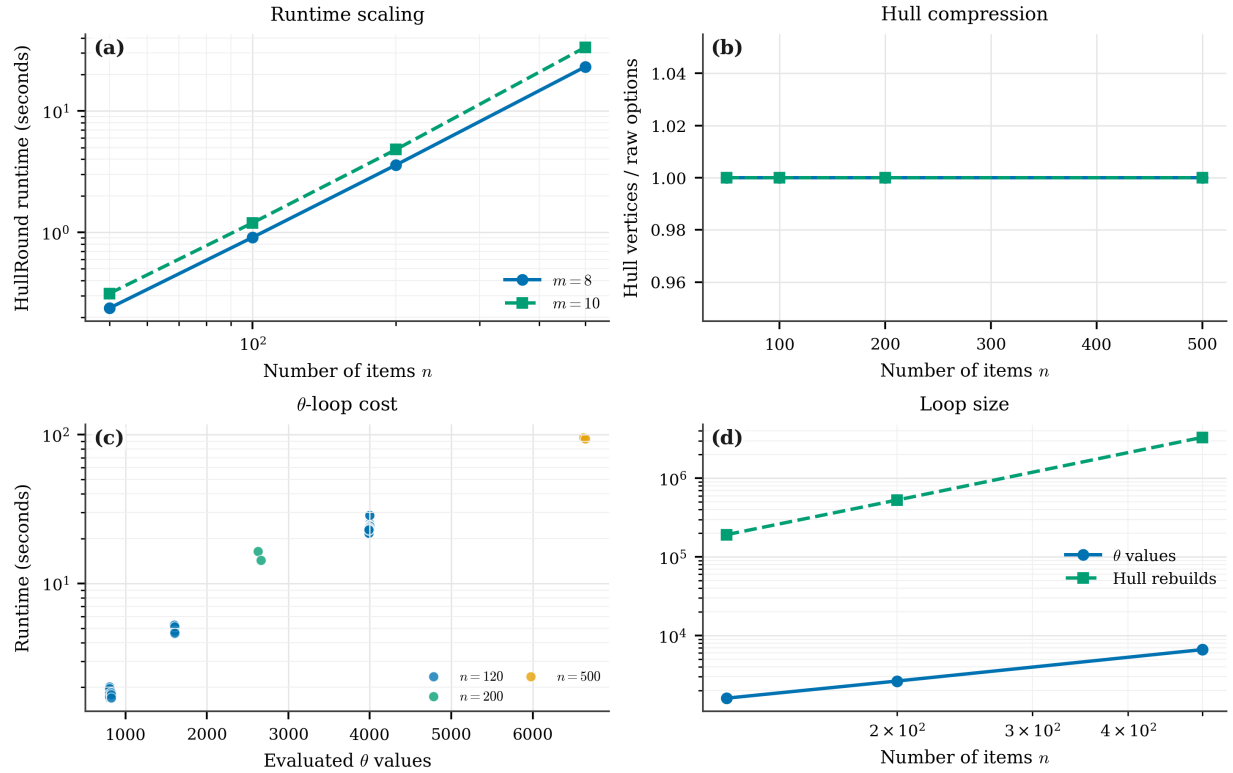


Figure C.4. Computational scaling diagnostics. Panels (a)–(b) summarize HullRound runtime and hull compression on 24 scaling rows; panels (c)–(d) show that runtime tracks the number of evaluated θ values and resulting hull rebuilds in the θ -loop diagnostics. The diagnostic motivates the exact parametric sweep in [Proposition 16](#); correctness still comes from the full original candidate set.

Table C.4. Family-level diagnostics for optimized exact θ -B&B. The HullRound gap column reports the median relative objective gap to the certified exact optimum where the exact optimum is available; it is omitted for families with no certified exact rows.

Family	Inst.	Cert.	Limited	Med. time (s)	Med. θ pruned	HullRound gap	Note
Economic	24	24	0	0.6476	90.6%	0.0032%	–
Hull compression	8	8	0	0.0047	100.0%	0.0000%	–
Low compression	8	8	0	0.0079	100.0%	0.0000%	–
Tight capacity	8	0	8	9.5959	0.0%	–	tight rows limit
Many θ	8	8	0	0.2175	100.0%	0.0000%	–
Boundary Γ	8	8	0	0.0088	100.0%	0.0000%	–

Table C.5. Targeted difficult-regime follow-up. The many- θ reduced diagnostic grid completed; the tight-capacity run was stopped after three instances because the per-method limits were being exhausted. The table is included to document robustness of the interpretation, not as a complete benchmark.

Family	Inst.	B&B cert.	B&B lim.	SCIP cert.	HiGHS cert.	B&B time	SCIP time	HiGHS time	Observation
Tight capacity	3	1	2	1	1	45.1174	18.8751	22.6098	limits with gaps; HR gap 0.408%
Many θ	24	24	0	24	24	2.6200	0.0111	0.0208	all certified; HR gap 0.000%

Table C.6. Safe maintenance diagnostics from the bounded construction ablation. Each family has eight construction cases. Dirty fraction is the median item-level share whose transformed point set changes; reuse rate is the median item-hull reuse share.

Family	Med. $ \mathcal{B} $	Dirty frac.	Reuse rate	Rebuilds	Reuses	Transform s	Hull s	LP s	Total s
many- θ	351	0.511	0.489	8975	8575	0.0170	0.6004	0.0432	0.7138
low compression	351	0.846	0.154	14851	2699	0.0179	0.6201	0.0501	0.7428
tight capacity	8.0	0.925	0.075	370	30	0.0003	0.0134	0.0012	0.0320
non-tight control	4.0	0.965	0.035	193	7	0.0001	0.0058	0.0004	0.0143

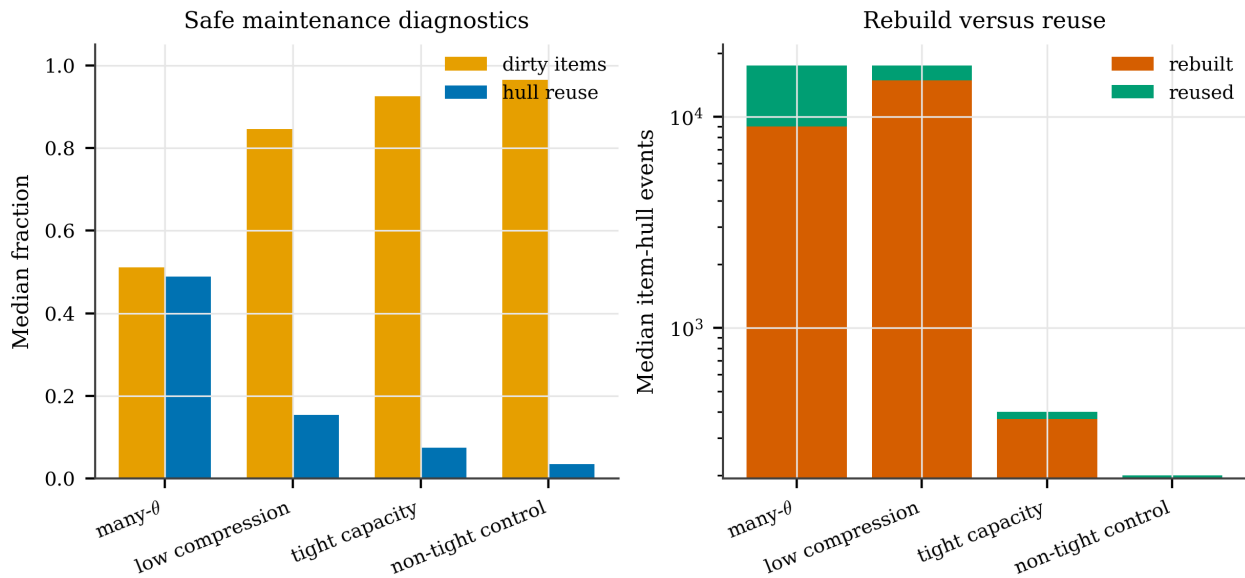


Figure C.5. Item-local maintenance diagnostics from the final sweep ablation. The left panel reports median dirty-item and reuse fractions; the right panel reports median item-hull rebuild and reuse events. The diagnostic shows where the safe maintenance rule avoids rebuilds and where frequent geometry changes leave little reusable state.

Table C.7. Matched solver comparison for the final sweep campaign. Times are medians in seconds on the four matched rows per family. HiGHS is the SciPy/HiGHS backend, SCIP is PySCIPOpt/SCIP, and Gurobi/CPLEX were unavailable in this environment.

Family	Matched	Enum s	Sweep s	Ratio	HiGHS s	SCIP s	Gurobi/CPLEX unavailable	Sweep cert.	Med. gap
many- θ	4	0.697	0.851	1.24	2.394	2.300	4/4	4/4	0.0000
low compression	4	0.793	0.996	1.19	3.355	2.789	4/4	4/4	0.0000
tight capacity	4	0.073	0.091	1.25	0.297	0.250	4/4	4/4	0.0000
non-tight control	4	0.011	0.019	1.75	0.007	0.012	4/4	4/4	0.0000

Table C.8. Non-tight HullRound quality stress test. Gaps are computed only against certified exact optima from optimized θ -B&B. The SCIP/HiGHS column reports certified rows for SCIP and HiGHS, respectively.

Family	Inst.	Exact cert.	HR feas.	Med. HR gap	Max HR gap	HR time	Exact time	SCIP/HiGHS cert.	Observation
Economic	16	16	16	0.00288%	0.00474%	0.4320	1.9654	16/16	near-exact
Many θ	16	16	16	0.00000%	0.00000%	0.4023	1.6611	16/16	exact match
Hull compression	16	16	16	0.00000%	0.00000%	0.0057	0.0131	16/16	exact match
Low compression	16	16	16	0.00000%	0.00000%	0.0084	0.0255	16/16	exact match

C.1 Claim-to-evidence summary

Table C.9. Claim-to-evidence summary for the main mathematical and computational claims. Limitations state where the evidence should not be extrapolated.

Claim	Evidence base	Exact status	Headline number	Limitation
Robust counterpart and θ enumeration	Proofs in Sections 2–6; validation tests	Mathematical proof; full $\mathcal{B}$ required	$\mathcal{B} = \{0\} \cup \{ t_{ij} \}$ over original admissible options	Finite menus; integer Γ ; positive robust denominator or residual primitive
HullRound quality	Non-tight stress-test benchmark	Robust-feasible heuristic compared only on certified exact rows	64/64 feasible; median gap 0; max gap 4.74×10^{-5}	Non-tight synthetic families; no exactness claim for HullRound
Exact solver certification	Certified benchmark	Exact when all nodes and θ subproblems close; valid gaps under limits	56/64 certified; 8 tight-capacity valid-gap rows; SCIP/HiGHS match certified rows	SCIP/HiGHS faster on median certified row in this environment
Parametric θ sweep	Final sweep validation and ablation	Exact when using the full breakpoint set and safe item-local hull reuse/rebuilds	16/16 enumeration-sweep matches; no certificate failures; median sweep/enumeration runtime ratios exceed one	Diagnostic and certification evidence; no broad runtime-superiority claim
Tight-capacity limitations	Targeted tight-capacity diagnostic	Limited rows are valid-gap diagnostics, not optima	Targeted tight capacity: 1/3 certified; limited gaps 9.4×10^{-4} and 1.48×10^{-3}	Residual LP bounds, weak cutoff pruning, branching, and full θ enumeration remain bottlenecks
Semi-synthetic pricing frontier	Application frontier and stress-test records	Robust certificates plus out-of-model stress tests; exact small-subset check	$\Gamma = 13$: revenue 0.9968, changed share 0.1167, IID violation 0; subset gap max 1.71×10^{-5}	No transaction-level scanner-data validation; no causal demand or cross-price substitution

C.2 Empirical synthesis

Figure C.6 gives a compact visual synthesis of the empirical evidence already tabulated above: HullRound as a robust-feasible primal method, exact θ -B&B as an independent certificate and valid-gap mechanism, and the semi-synthetic application as a pricing-decision frontier rather than a runtime benchmark.

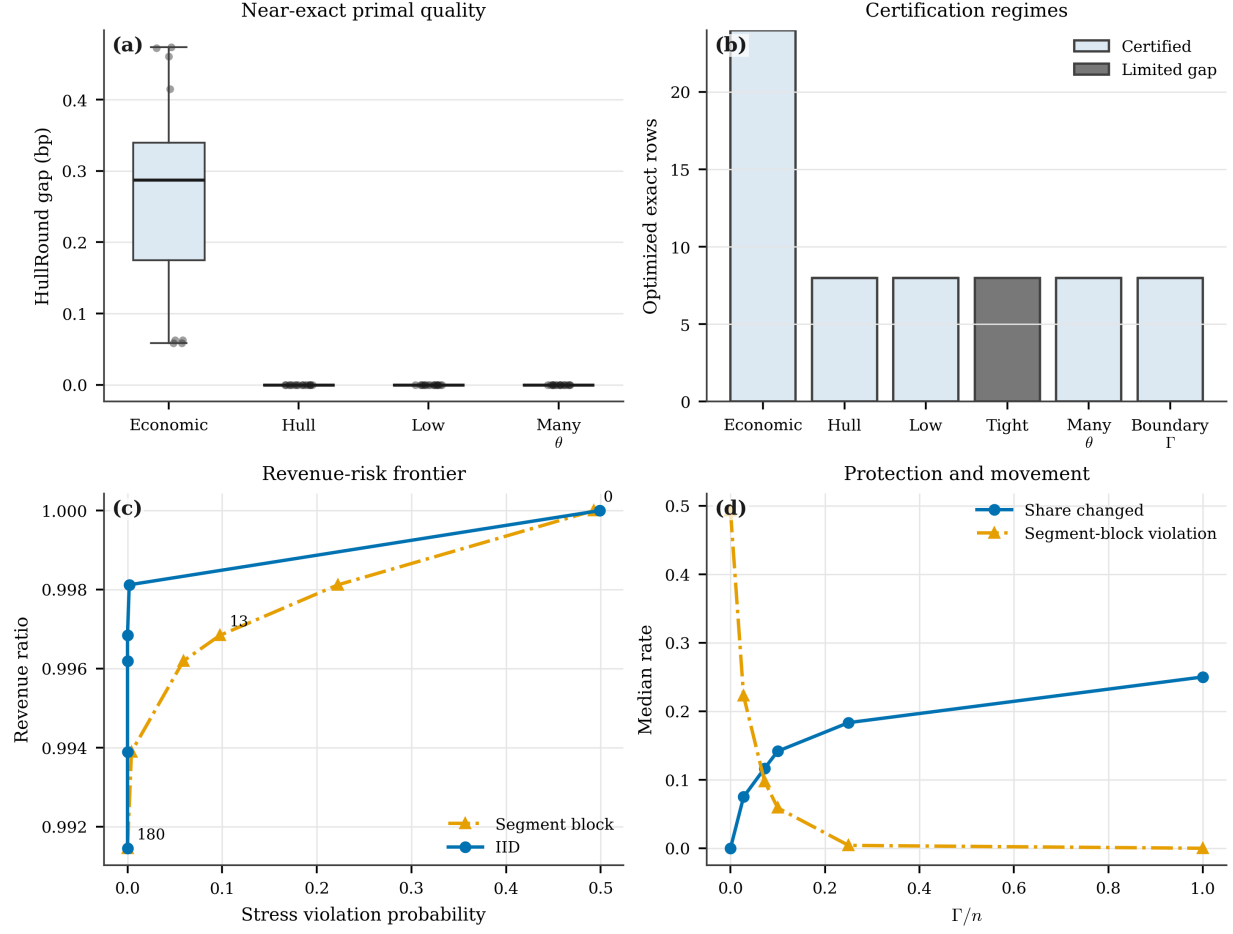


Figure C.6. Empirical synthesis. Panel (a) reports HullRound gaps to certified exact optima in the non-tight stress test. Panel (b) shows optimized exact θ -B&B status by family in the certified benchmark. Panels (c)–(d) summarize the semi-synthetic revenue–risk and price-movement frontier.

D Semi-Synthetic Pricing Diagnostics

Table D.1. Segment diagnostics at the moderate protection level. Price changes are relative to each seed’s nominal policy; uncertainty is the segment-level proportional demand uncertainty scale used by the generator.

Segment	Median Δp	Share changed	Uncert.	Δ margin
mainstream	0.000	0.083	0.140	173.9
premium	0.000	0.345	0.120	38.7
private_label	0.000	0.000	0.220	0.0
seasonal	0.000	0.036	0.300	0.5
staples	0.000	0.098	0.080	364.6

Table D.2. Exact small-subset quality check for the semi-synthetic application generator. The exact rows use $n = 50$, $m = 10$, two seeds, and the moderate Γ values from the subset; gaps are reported only when exact global θ -B&B certifies optimality.

Method	Exact cert.	Feasible	Med. gap	Max gap	Med. time
Global θ -B&B	4	4	0.000000	0.000000	1.222
HullRound	0	4	0.000012	0.000017	0.236

E Reproducibility Details

The public repository contains the solver package, tests, experiment drivers, plotting and summarization scripts, and reproducibility instructions. The exact manuscript-generation snapshot for this submission uses generated auto-number files, tables, figures, and result summaries from the submission artifact bundle. The lightweight validation command is `python -m pytest -q` in an environment with the package dependencies installed. The manuscript builds in this environment with `tectonic main_v2.tex` from `paper_versions/v2`; the generated manuscript artifacts are read from the local `auto/`, `tables/`, and `figures/` directories. Full benchmark runtimes depend on hardware and solver availability. SCIP and HiGHS versions used for the reported benchmark are recorded in the generated result summaries; commercial solvers were unavailable for the reported runs.